FPT ACADEMY INTERNATIONAL

FPT – APTECH COMPUTER EDUCATION

InterVue

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Semester: *IV*

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InterVue

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Acknowledge

Every software project begins with a simple question. Ours was this: *why is it still so hard for a talented developer and a company that needs them to find each other and trust each other?* InterVue grew out of that question. As we bring this project to completion, we feel a deep sense of accomplishment, and an even deeper sense of gratitude to everyone who helped turn that question into a working platform.

InterVue was never meant to be another job board. We set out to build a recruitment platform dedicated to the Information Technology industry, where every job post states its salary, every business is verified before it reaches candidates, every candidate confirms their identity before applying, and every step of the hiring process can be seen and followed. Translating that vision into software turned out to be far more demanding than we first imagined, and far more educational.

A Journey Through Real-World Complexity

This project placed us in front of problems that rarely appear in textbooks, but appear in every real product.

We learned that **trust has to be engineered**. Building electronic identity verification with FPT.AI meant much more than calling an API. We had to decide what happens when a confidence score falls into the uncertain range, when the same citizen ID appears on two accounts, or when a user fails verification five times. On the employer side, we designed a three-level verification process, from email to business license to domain verification, so that candidates can tell a genuine company from a fraudulent one at a glance.

We learned that **money demands precision**. Integrating bank transfers through SePay taught us to think in terms of orders, expiry windows, idempotent processing, and exact reconciliation. A single transaction must never be credited twice, and an

underpayment must never silently activate a service. These details may be invisible to users, but they decide whether a platform deserves its users' trust.

We learned that **security is a series of careful decisions**. Generic login error messages, CAPTCHA after repeated failures, two-factor authentication with recovery codes, single-use links that expire in minutes, and a seven-day grace period before an account is permanently deleted are not glamorous features. Each one, however, reflects a decision about how to protect people's accounts and personal data.

We learned that **real-time systems change how people communicate**. Implementing chat over WebSocket, interview scheduling with reminders, and a clear application timeline showed us how much a candidate's experience depends on knowing where they stand. A simple "Viewed" status, with the time a recruiter opened their CV, can mean a great deal to someone waiting for an answer.

Most importantly, we learned that **artificial intelligence must be used responsibly, especially when people's careers are at stake**. Throughout InterVue, AI evaluates CVs, simulates interviews, recommends jobs, drafts job descriptions, and answers usage questions through a retrieval-augmented chatbot. From the very beginning, however, we committed to a clear principle: *AI assists, humans decide*. Every AI assessment must point to evidence in the CV. AI must never invent experience or skills a candidate does not have. It must never use age, gender, photos, hometown, or identity data to judge a person. A missing detail in a CV does not mean a missing skill. Holding ourselves to these principles made us think not only as developers, but as people responsible for the impact of what we build.

Each of these challenges forced us to slow down, ask harder questions, and accept that good software is defined as much by its rules, limits, and edge cases as by its visible features.

Heartfelt Gratitude to Our Supervisor and FPT Aptech

First and foremost, we would like to express our sincere gratitude to **FPT Aptech** for giving us the opportunity to undertake this project. The project-based learning

approach allowed us to experience the full lifecycle of software development, from problem definition and requirement analysis to system design, implementation, and testing, in a way that no lecture alone could provide.

We are especially grateful to our supervisor, **Mr. HOÀNG ĐỨC QUANG**. His guidance has been the compass of this project. He taught us to define requirements precisely before writing a single line of code, to question our own assumptions, and to examine every feature from the perspective of each party who would use it: the candidate, the recruiter, and the administrator. His thoughtful feedback during each review pointed out weaknesses we had overlooked and pushed our work to a higher standard. Beyond technical knowledge, he showed us the mindset of a professional engineer: careful, honest, and always focused on the real problem. Without his patience and dedication, InterVue would not have reached the level it has today.

A Salute to Our Team

This project belongs to the four of us: **Phạm Hoàng Tuấn, Mai Trung Hậu, Lê Minh Trí, and Lâm Hoàng An**. Along the way we shared long discussions over requirements, late nights of debugging, disagreements that were settled by better arguments rather than louder voices, and the quiet satisfaction of watching separate modules finally come together as one system.

Working together taught us lessons that go beyond code. We learned to divide responsibilities clearly, to review each other's work with honesty and respect, to communicate when something was going wrong, and to hold the whole system to a shared standard rather than to individual comfort. Each member brought different strengths, and InterVue is stronger because of all of them. We are proud of what we built together, and even prouder of how we grew as a team in the process.

A Heartfelt Thanks to Our Families and Friends

We also owe deep thanks to our families and friends. They patiently accepted our long hours in front of screens, our missed gatherings, and our constant talk of deadlines and bugs. Their encouragement kept us going through the most demanding

stages of this project, and their belief in us gave us the confidence to aim higher. Though their names do not appear in our source code, their support is present in every part of this work.

A Project That Transcends the Classroom

For us, InterVue is far more than an academic requirement. It reflects a problem we genuinely care about. Many young developers, interns, and fresh graduates struggle to find opportunities they can trust, while many companies struggle to identify the right people among hundreds of applications. Through InterVue, we have tried to show that technology can make this process clearer, fairer, and more humane: salaries are transparent, businesses and candidates are verified, progress can be tracked at every stage, and AI supports human judgment instead of replacing it.

Looking Forward to New Horizons

As we close this chapter, we carry with us much more than a finished project. We carry the experience of analyzing real requirements, designing systems that must handle failure gracefully, protecting users' data and money, and using artificial intelligence with care and accountability. These lessons will remain with us long after this semester ends, and they will guide us as we step into the IT industry as developers, problem-solvers, and responsible builders of technology.

With heartfelt thanks and great excitement for the journey ahead,

Team 01

Synopsis

InterVue is a recruitment platform dedicated to the Information Technology industry. It connects IT job seekers, from interns and fresh graduates to senior engineers, with businesses that need technical talent. The platform covers the entire hiring journey in one place: publishing job posts, discovering opportunities, submitting applications, screening candidates with the support of artificial intelligence, scheduling interviews, communicating in real time, and recording the final hiring outcome. Its defining principle is simple: *AI assists, humans decide.*

Background and Motivation

The IT job market in Vietnam is growing rapidly, yet the hiring experience for both sides remains frustrating. Candidates frequently encounter job posts with vague requirements, hidden salaries, or fraudulent intentions, including requests to pay deposits or fees before being considered. Once they apply, many receive no feedback at all and have no way of knowing whether their CV has even been opened. Recruiters, on the other hand, spend a large amount of time manually reading hundreds of applications, many of which do not match the position, and they have few reliable tools to compare a CV against the actual technical requirements of a job. On general-purpose platforms, there is often little to confirm that a candidate is who they claim to be, or that a company posting a job is a genuine, legally registered business.

InterVue addresses these problems through four commitments: **transparency** of job information, **trust** between both parties, **efficiency** in evaluating applications, and **visibility** of the application process.

Users and Roles

The system serves three roles.

Candidates are people looking for full-time, part-time, or internship positions in IT. **Recruiters** are representatives of businesses or household businesses who publish job posts and manage hiring; individuals cannot post jobs on InterVue. **Administrators** operate the platform: they verify identities and businesses, moderate job posts and reports, manage shared catalogs and service packages, handle transactions and refunds, and respond to support requests.

Candidate and recruiter accounts are maintained through separate portals. The same email address may be used to register on both, but each account has its own credentials, profile, verification status, service packages, and transaction history. A recruiter can never apply to a job published under their own account.

Building Trust on Both Sides

Trust is the foundation of InterVue, and it is enforced through verification rather than assumed.

Before submitting any application, a candidate must complete electronic identity verification (eKYC) through FPT.AI. The process consists of capturing both sides of the citizen identity card and performing a liveness face check. High-confidence results are approved automatically. Uncertain results, duplicate identity numbers, or repeated failures are escalated to an administrator for manual review. The minimum age of 16 is confirmed from the date of birth on the identity card.

Recruiters progress through three verification levels. The first level requires a verified email. The second requires phone verification, complete company information matched against tax records, and an approved business or household-business license. The third level adds verification through an email address on the company's own domain. Each level unlocks additional capabilities, such as free job posts, CV views in candidate search, and a visible verification badge on job posts and company pages. Recruiters must also accept data processing agreements before they can receive any CV.

Every job post, without exception, is reviewed by an administrator before it becomes visible. AI pre-screens each post for warning signs such as requests for payment from candidates, discriminatory requirements, or abnormal salaries, and flagged posts are moved to the front of the review queue. Candidates can report suspicious posts. When three different verified candidates report the same post within 24 hours, it is automatically hidden until an administrator investigates.

Transparent Job Information

Every job on InterVue is paid; unpaid positions are not permitted. Each post states the position, level, required skills, working arrangement (on-site, hybrid, remote, or flexible), contract type, work location, and salary. The salary can be shown as a range, a minimum, a maximum, or as negotiable. For full-time and part-time positions, the system validates that the stated salary is not below the regional minimum wage defined by Vietnamese law.

Positions, levels, skills, and locations are drawn from standardized catalogs shared by both sides. Locations follow Vietnam's new two-level administrative structure of province or city and ward or commune. This shared vocabulary allows job search, recommendations, and AI matching to compare candidates and jobs consistently.

The Candidate Experience

Candidates can search for jobs by title, company, and location without signing in, and can explore jobs near them on a map within a chosen radius. After registering, a guided onboarding helps them upload a CV, let AI fill in their profile, confirm their desired position, and enable job alerts.

The candidate profile goes well beyond a single document. It includes work experience, standardized skills with proficiency levels and years of use, notable projects with links to source code and live demos, education, certificates, languages, and job expectations. Candidates control what recruiters can see. Birth year and gender are hidden by default and are never used by AI, and contact details are revealed only to

recruiters of jobs the candidate has applied to, or after the candidate accepts a chat invitation.

Once applications are submitted, candidates follow each one through a clear timeline of statuses: Submitted, Viewed, Under Review, Interview, Hired, Not Selected, Withdrawn, or Job Closed. They can see exactly when a recruiter first opened their CV. They can confirm, decline, or propose a new time for interviews, add them to Google Calendar, and receive reminders 24 hours and one hour in advance.

The Recruiter Experience

Recruiters organize their hiring into campaigns, where each campaign represents one position and contains its job post, applications, recommended candidates, active services, and reports. AI helps them draft job descriptions from a position and level, or fill in a post automatically from an existing job description file.

Recruiters can request that candidates answer screening questions, set expected answers, and have non-matching applications flagged rather than rejected. They manage applications through a structured pipeline, keep internal notes, select standardized rejection reasons, and choose whether to share those reasons with the candidate. Verified recruiters can also search for candidates who have chosen to be discoverable, view AI-recommended profiles for each campaign, and invite candidates to chat. Recruitment reports present the hiring funnel, from job views to applications, interviews, and hires, along with response rates and the effect of paid services.

Artificial Intelligence with Accountability

AI is integrated throughout InterVue, always as an assistant and never as a decision-maker.

For candidates, AI reviews CVs at three levels: a general review against common standards, a review against the candidate’s desired position, and a detailed fit check against a specific job post. It also provides text-based mock interviews with feedback on each answer, job recommendations that explain why each job fits, and cover letter

drafts based on the candidate's own CV. For recruiters, AI compares each CV against the job requirements, highlights matched and missing skills with evidence from the CV, classifies applications into high, medium, or low fit, and points out what should be clarified in an interview. A separate chatbot built on retrieval-augmented generation answers questions about how to use the platform, and only about the platform.

These features follow strict principles. Every assessment must be grounded in evidence from the CV, presented as the relevant passage, the issue, and a suggested fix. AI must never invent experience or skills the candidate does not have. Age, gender, photos, hometown, and identity data are never used in any evaluation. A missing detail in a CV is never interpreted as a missing skill. AI never rejects or hires anyone automatically, and users are informed before their data is processed by AI. If an AI request fails, no quota is deducted, and paid usage is refunded automatically.

Security, Payments, and Data Protection

InterVue supports registration with email or social accounts, enforces strong password rules, and requires CAPTCHA after repeated failed login attempts without locking accounts. Users can enable two-factor authentication through an authenticator app or email, keep recovery codes, review their active devices, and sign out of all devices at once.

Paid services, such as candidate plans, job-fit checks, promoted job posts, add-ons, and recruiter credits, are purchased through bank transfer via SePay using a QR code generated for each order. The system automatically matches incoming transactions to orders, processes each transaction exactly once, and activates the service only when the payment is sufficient.

Account deletion follows a seven-day grace period, after which personal data, identity verification data, CVs, and conversations are removed. Only transaction records required by accounting and tax regulations, and records related to violations, are retained.

Platform and Scope

InterVue is designed primarily as a responsive web application, which offers the most comfortable environment for reading and comparing CVs, managing job posts, and operating the system. Its long-term direction includes a mobile application built with Flutter for quick job discovery, application tracking, and notifications, sharing the same accounts, job posts, and applications as the web platform.

Conclusion

InterVue brings together verified identities, transparent job information, a visible application pipeline, and responsible artificial intelligence in a single platform built for the IT industry. For candidates, it offers opportunities they can trust and a clear view of where they stand. For recruiters, it offers faster and better-informed screening without giving up human judgment. For our team, it has been a practical exercise in the full software development lifecycle, from defining the problem and analyzing requirements to designing a secure, reliable, and fair system that addresses a real need in the market.

Review 01

Functional Requirements

I. Functional Requirements

This section describes what each type of user can do in InterVue. The system has four roles: guest, candidate, recruiter, and administrator. A guest is a visitor who has not signed in. Candidate and recruiter accounts are separate, even when they use the same email address, and a Super Admin creates administrator accounts.

1. Guest

1.1 Browsing jobs and companies

- A guest can search for jobs by job title, company name, or location.
- A guest can read the details of a job post.
- A guest can view nearby job posts on the "Jobs near you" map.
- A guest can choose a search radius on the map.
- A guest can view a company page and its current openings.
- A guest can view the featured companies on the homepage.
- A guest can read the FAQ and Blog pages.

1.2 Using the help chatbot

- A guest can ask the help chatbot questions about how to use InterVue.

1.3 Registering and signing in

- A guest can register a candidate account with an email address or a social media account.
- A guest can register a recruiter account with an email address or a social media account.
- A guest can sign in with an email address and password or a social media account.
- A guest can request a password reset link by entering the account email address.

2. Candidate

2.1 Account verification and security

- A candidate can verify an email address by opening the verification link sent to that address.
- A candidate can request a new verification email.
- A candidate can complete identity verification (eKYC) with a citizen identity card and a face check.
- A candidate can retry identity verification and can request a manual review.
- A candidate can enable two-factor authentication.
- A candidate can review the active devices and sign out of one device or of all devices.

- A candidate can link or unlink social accounts and can change the password or the email address.

2.2 Onboarding and profile

- A candidate can follow a guided onboarding to set up the profile, and can skip steps and resume them later.
- A candidate can view a dashboard that summarizes applications, interviews, recommended jobs, and profile status.
- A candidate can manage a profile with personal information, experience, skills, education, and job expectations.
- A candidate can control which personal information recruiters can see and whether recruiters can find the candidate.
- A candidate can review the profile changes that AI proposes from a newly uploaded CV and choose which ones to apply.

2.3 CV management

- A candidate can upload a CV as a PDF file.
- A candidate can rename a CV, set a default CV, and delete a CV.
- A candidate can store several CVs, depending on the service plan.

2.4 Job discovery

- A candidate can filter job search results by position, level, salary, and working arrangement.

- A candidate can save jobs and review them later.
- A candidate can receive AI job recommendations that explain why each job fits.
- A candidate can create daily or weekly job alerts.
- A candidate can follow companies and receive notifications about their new job posts.

2.5 AI support

- A candidate can request AI feedback on a CV in general, for the desired position, or for a specific job post.
- A candidate can practice text-based mock interviews and receive feedback on each answer and a summary.
- A candidate can review the history of past mock interview sessions.
- A candidate can use AI to draft a cover letter, then edit, save, and download it as a PDF file.

2.6 Applications

- A candidate can apply for a job with a CV and answers to screening questions.
- A candidate can follow the status of each application.
- A candidate can withdraw an application and can reapply later.

2.7 Interviews and chat

- A candidate can confirm, decline, or propose a new time for an interview.

- A candidate can add an interview to Google Calendar and receive reminders before it starts.
- A candidate can chat in real time with recruiters.
- A candidate can accept or decline a chat invitation from a recruiter.

2.8 Plans, support, and safety

- A candidate can purchase service plans and review the payment history.
- A candidate can submit support requests.
- A candidate can report suspicious job posts and can block companies.
- A candidate can request account deletion.

3. Recruiter

3.1 Registration and verification

- A recruiter can post jobs on behalf of a business or a household business.
- A recruiter can describe the current hiring needs to receive suitable service suggestions.
- A recruiter can verify an email address (Level 1), and can then prepare the account, create campaigns, and draft job posts.
- A recruiter can verify a phone number with an OTP.
- A recruiter can complete the company profile and can submit a business license for approval.

- A recruiter at Level 2 can post jobs for free, view CVs in candidate search, and display a verification badge.
- A recruiter can verify an email address on the company's own domain (Level 3) and receive a high-level verification badge.
- A recruiter can accept the data processing agreements.
- A recruiter can enable two-factor authentication.

3.2 Company management

- A recruiter can select an existing company or create a new company.
- A recruiter can submit a letter of introduction or authorization to join an existing company.
- A recruiter can request changes to locked company information with supporting documents.

3.3 Campaigns and job posts

- A recruiter can view a dashboard that summarizes campaigns, job posts, applications, interviews, services, and warnings.
- A recruiter can create a recruitment campaign for each position.
- A recruiter can create a job post with structured information such as position, level, skills, and work location.
- A recruiter can state a salary for each job post.

- A recruiter can use AI to draft a job description or to fill in a job post from an existing description file.
- A recruiter can configure screening questions with expected answers.
- A recruiter can submit job posts for review and can edit, extend, pause, reopen, and close them.
- A recruiter can edit a live job post while the previously approved version stays visible until the change is approved.

3.4 Application management

- A recruiter can filter, search, and sort the applications of a campaign.
- A recruiter can view CVs, screening answers, and the AI fit assessment of each application.
- A recruiter can change the status of an application and record a rejection reason.
- A recruiter can keep internal notes on an application.
- A recruiter can save promising candidates and can report suspicious candidates.

3.5 Candidate search

- A recruiter can search for candidates who have chosen to be discoverable.
- A recruiter can view the AI-recommended profiles of each campaign by using CV view quotas or credits.
- A recruiter can send a chat invitation to a candidate.

3.6 Interviews and chat

- A recruiter can create an interview schedule for an application.
- A recruiter can accept a new time proposed by a candidate or propose another time.
- A recruiter can chat in real time with candidates.

3.7 Services, payments, and reports

- A recruiter can purchase service packages, add-ons, and credits.
- A recruiter can activate a purchased service on a job post.
- A recruiter can request a VAT invoice for a payment.
- A recruiter can review the history of services, credits, and account changes.
- A recruiter can view recruitment reports on the funnel, processing time, candidate sources, and service effectiveness.

4. Administrator

4.1 Access and overview

- A Super Admin can create administrator accounts.
- An administrator can sign in with two-factor authentication.
- An administrator can access the modules of the assigned internal sub-role.
- An administrator can view an overview dashboard of users, pending verifications, pending job posts, reports, and revenue.

- An administrator can view the audit log of administrator actions.

4.2 Accounts and verification

- An administrator can search for, view, lock, and unlock candidate and recruiter accounts.
- An administrator can reset the two-factor authentication of a user.
- An administrator can review escalated eKYC cases and can approve, reject, or grant additional attempts.
- An administrator can review business verification requests and company information changes, and can approve, reject, or request additional documents.

4.3 Content moderation

- An administrator can review job posts in a prioritized queue.
- An administrator can compare an edited job post with its approved version.
- An administrator can approve, reject, or hide job posts.
- An administrator can process reports on job posts, candidates, and conversations, and can inform the reporter of the outcome.

4.4 Catalogs, services, and finance

- An administrator can maintain the catalogs of positions, levels, and skills.
- An administrator can maintain the administrative units and the regional minimum wage table.

- An administrator can configure service plans, packages, add-ons, and credits.
- An administrator can monitor orders and transactions and can reconcile them manually when needed.
- An administrator can process refunds and VAT invoice requests and can export revenue reports.
- An administrator can generate, export, track, and revoke batches of Education plan codes.

4.5 Support and content

- An administrator can handle support tickets and consultation requests.
- An administrator can manage the FAQ, the Blog, the featured companies, and the knowledge base of the help chatbot.
- An administrator can send announcements to candidates, recruiters, or all users.

System Requirements

II. System Requirements

1. Server requirements

1.1 Hardware

Component	Requirement
CPU	**Processor type:** 64-bit (x86_64) multi-core processor **Processor cores:** Minimum: 2 cores · Recommended: 4 cores or more **Processor speed:** Recommended: 2.5 GHz or faster
OS	Ubuntu Server 22.04 LTS or higher (recommended) Windows Server 2019 or higher
Memory (RAM)	**Minimum:** 4 GB **Recommended:** 8 GB or more
Hard Drive	**Free space:** Minimum: 20 GB (SSD) · Recommended: 100 GB SSD or more **Maximum:** Operating system maximum
Network	Static public IP or domain name, with HTTPS (SSL certificate)

1.2 Software

Component	Requirement
Java Runtime	JDK 25
Backend Framework	Spring Boot (compatible with JDK 25)
Frontend Runtime	Node.js (LTS version supported by Next.js 16)
Frontend Framework	Next.js 16.x
RDBMS	MySQL 8.x
Cache / In-memory Store	Redis 7.x
Web Server / Reverse Proxy	Nginx (HTTPS, WebSocket support)
Third-party Services	FPT.AI (eKYC), SePay (payment), AI language model API, email delivery service

2. Client requirements

1.1 Hardware (Web)

Component	Requirement
CPU	**Processor type:** Intel Core i3 / AMD Ryzen 3 or equivalent **Processor speed:** Recommended: 2.0 GHz or faster
OS	Windows 10 or higher, macOS 12 or higher, Linux

Memory (RAM)	**Minimum:** 4 GB **Recommended:** 8 GB or more **Maximum:** Operating system maximum
Hard Drive	**Free space:** Minimum: 500 MB (browser cache and downloaded files)
Camera	Webcam or phone camera (required for identity verification)
Network	Stable Internet connection (minimum 5 Mbps)

1.2 Hardware (Mobile)

Component	**Requirement**
OS	Android 8.0 or higher
Memory (RAM)	**Minimum:** 3 GB **Recommended:** 4 GB or more
Storage	**Free space:** Minimum: 200 MB
Camera	Front and rear camera (required for identity verification)

Software (Mobile)

Component	**Requirement**
Mobile Application	InterVue mobile app (Flutter) for Android

3. Development Software

- IntelliJ IDEA (Backend – Spring Boot)

- Visual Studio Code (Frontend – Next.js, TypeScript, Tailwind CSS)
- Android Studio (Mobile – Flutter, Android build)
- JDK 25
- Node.js
- Flutter SDK
- MySQL 8.x and MySQL Workbench
- Redis
- Postman (API testing)
- Git and GitHub (version control)

III. Task Sheet – Review 1

S.No	Task	Deliverable	Assigned to	Status
1	Set up the project architecture, database, and security framework	Spring Boot and Next.js project base, database schema, JWT role-based authorization	Phạm Hoàng Tuấn	In Progress
2	Develop the AI module	AI CV evaluation, CV feedback, profile auto-fill, job recommendation, and job description generation	Phạm Hoàng Tuấn	Not Started
3	Develop the payment and service module	SePay payment, order reconciliation, service packages, and credit management	Phạm Hoàng Tuấn	Not Started
4	Integrate and deploy the system	Integrated system deployed on the server	Phạm Hoàng Tuấn	Not Started
5	Develop recruiter verification and company management	Phone verification, company profile, business license verification	Mai Trung Hậu	Not Started
6	Develop campaign and job post management	Campaign and job post CRUD, salary validation, screening questions, job post moderation	Mai Trung Hậu	Not Started
7	Develop application management and interview scheduling	Application pipeline, candidate search, interview schedules	Mai Trung Hậu	Not Started
8	Develop real-time chat and notifications	WebSocket chat, chat invitations, in-app and email notifications	Mai Trung Hậu	Not Started
9	Develop candidate profile and CV management	Candidate profile, CV upload and management, onboarding	Lâm Hoàng An	Not Started
10	Develop identity	eKYC integration and admin eKYC	Lâm Hoàng An	Not

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 01	hoa
Signature		
Date		

	verification (eKYC)	review		Started
11	Develop job search and job application	Job search, job map, saved jobs, job alerts, job application, application tracking	Lâm Hoàng An	Not Started
12	Develop authentication	Register, login, logout, email verification, password reset	Lê Minh Trí	In Progress
13	Develop guest pages	Home page, company page, FAQ, Blog	Lê Minh Trí	Not Started
14	Develop basic admin management	User management, catalog management, FAQ and Blog management	Lê Minh Trí	Not Started

Review 02

I. Architecture Design

1. Application Technology Architecture & Core Modules

The technical architecture integrates a Next.js 16 App Router web frontend, Flutter cross-platform mobile client, Spring Boot 4.1 reactive core with Java 25 virtual threads, and specialized microservices for AI processing and payment webhooks.

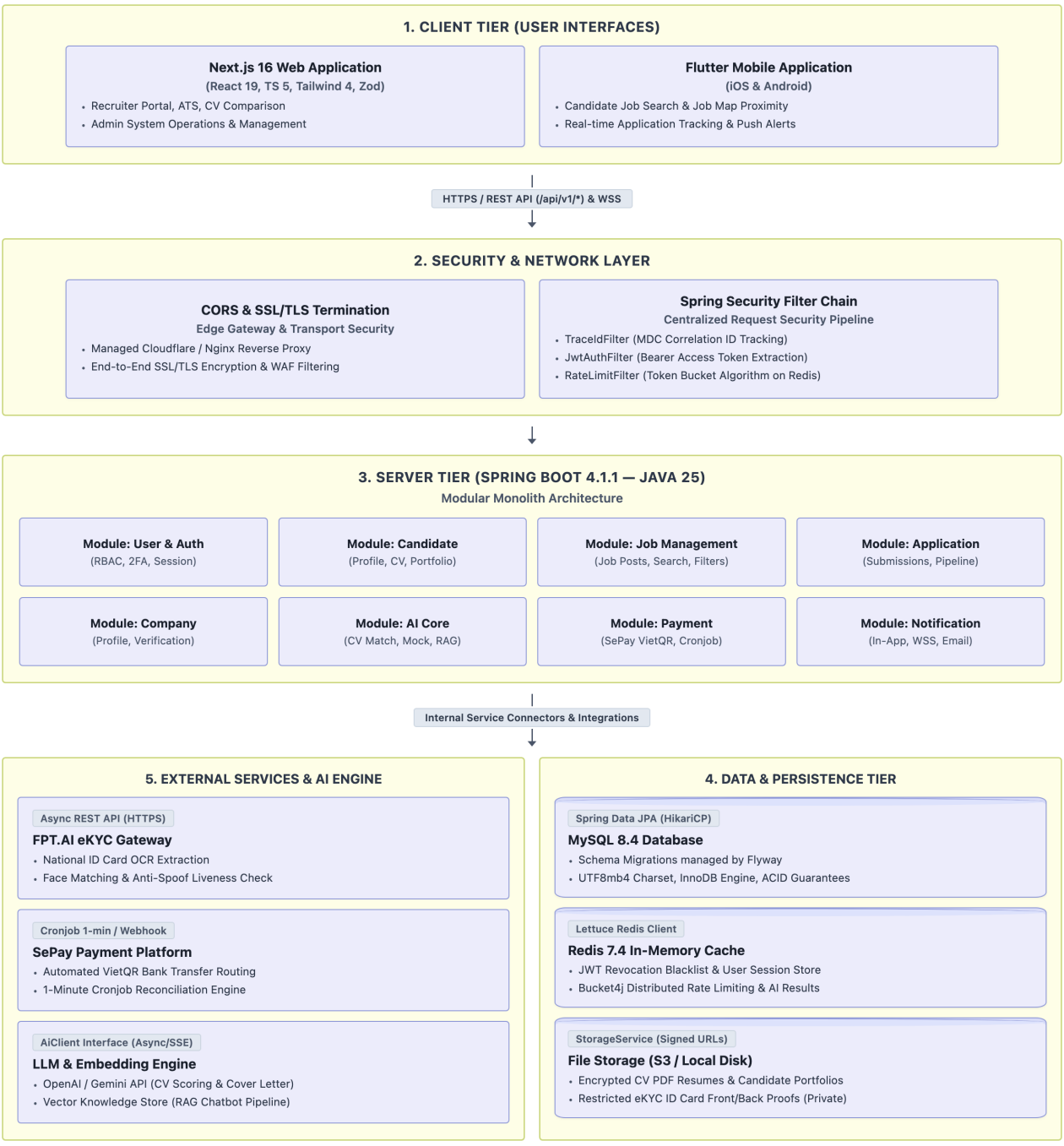


Figure 1.1. Overall Application Technology Architecture of InterVue Platform

2. Physical Deployment Architecture

The deployment architecture models the physical infrastructure topology, including AWS ALB ingress, Dockerized ECS containers across dual availability zones, PlanetScale MySQL replication, Redis cluster, and external SaaS gateways.

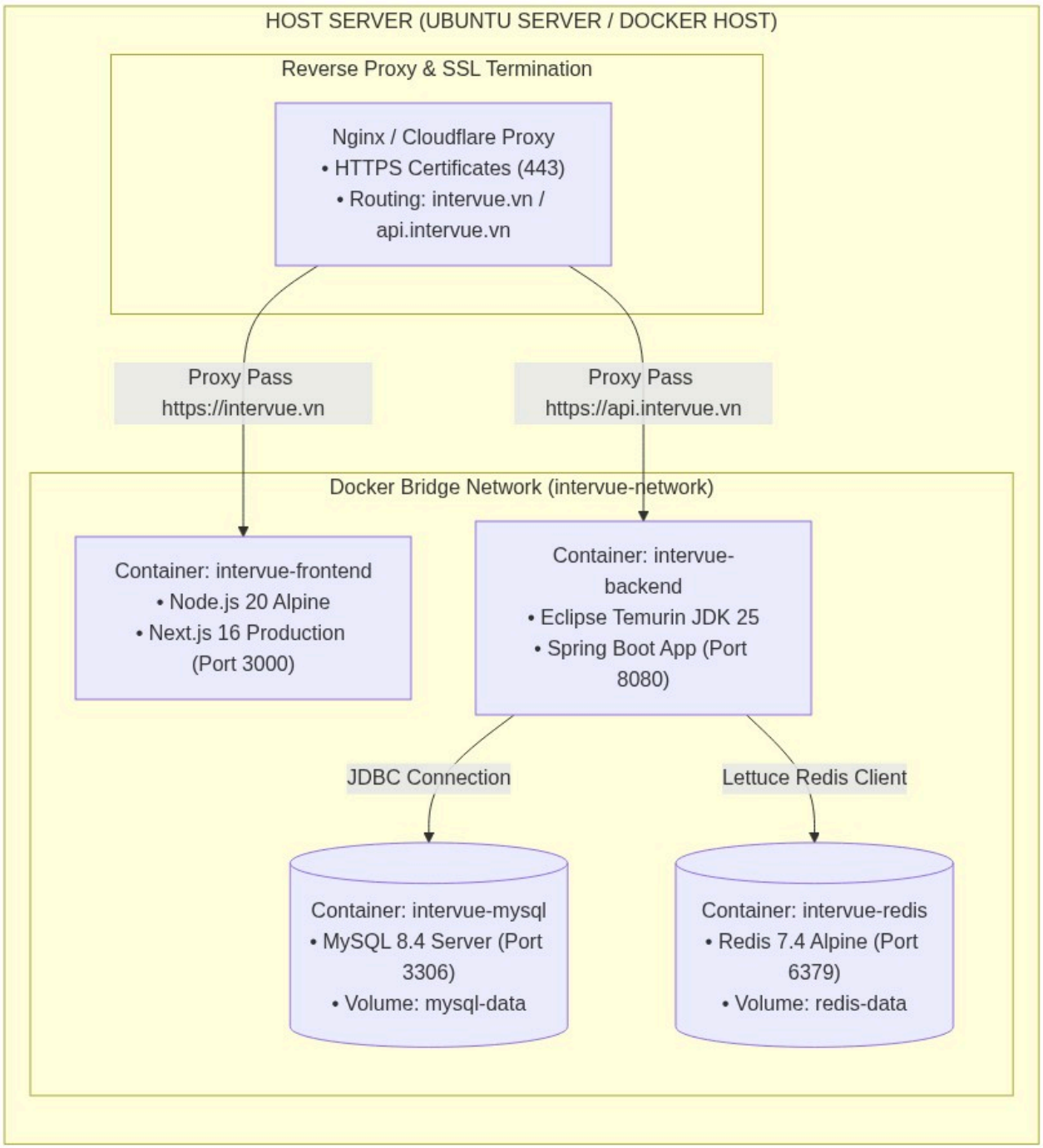


Figure 1.4. InterVue Physical Infrastructure & Cloud Deployment Architecture

II. Use Cases Diagrams

1. System Use Case Overview

The InterVue system boundary organizes business capabilities across four primary user roles: Guests, Candidates, Recruiters, and Administrators (including Super Admin). The platform coordinates identity verification, recruitment operations, AI-assisted mock interviews, and automated VietQR payment reconciliation.

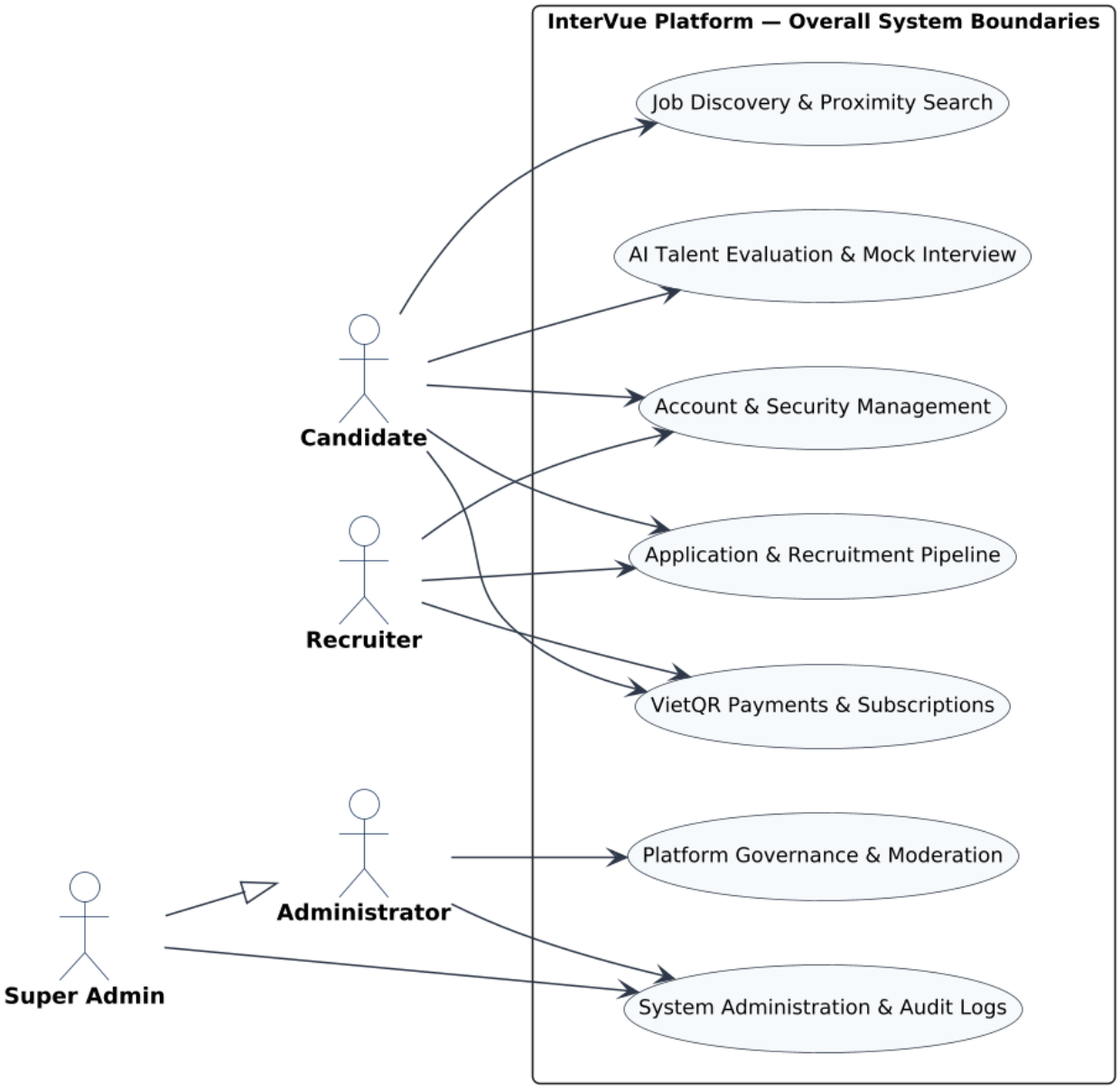


Figure 2.1. Overall System Use Case Overview of InterVue Platform

2. Candidate Subsystem Use Case Model

The Candidate Subsystem manages the end-to-end job seeker journey. Applying for jobs strictly requires verified eKYC status and attachment of a validated CV, with optional extensions for AI-generated cover letters and screening questionnaires.

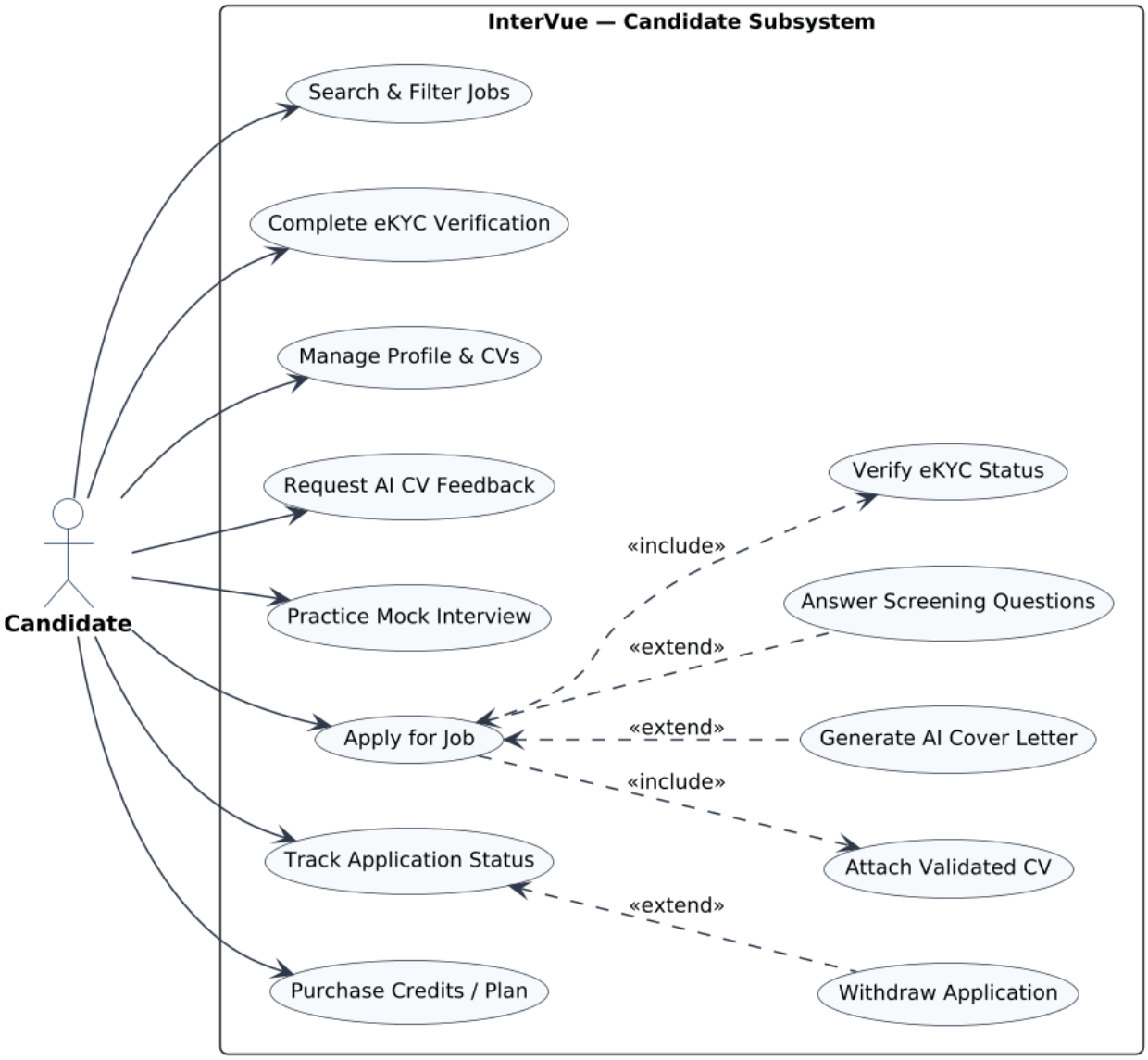


Figure 2.2. Candidate Subsystem Use Case Diagram

3. Recruiter Subsystem Use Case Model

The Recruiter Subsystem equips enterprise employers with tools for company profile verification, job posting lifecycle management, AI-driven candidate match fit evaluation, and SePay VietQR payment integration.

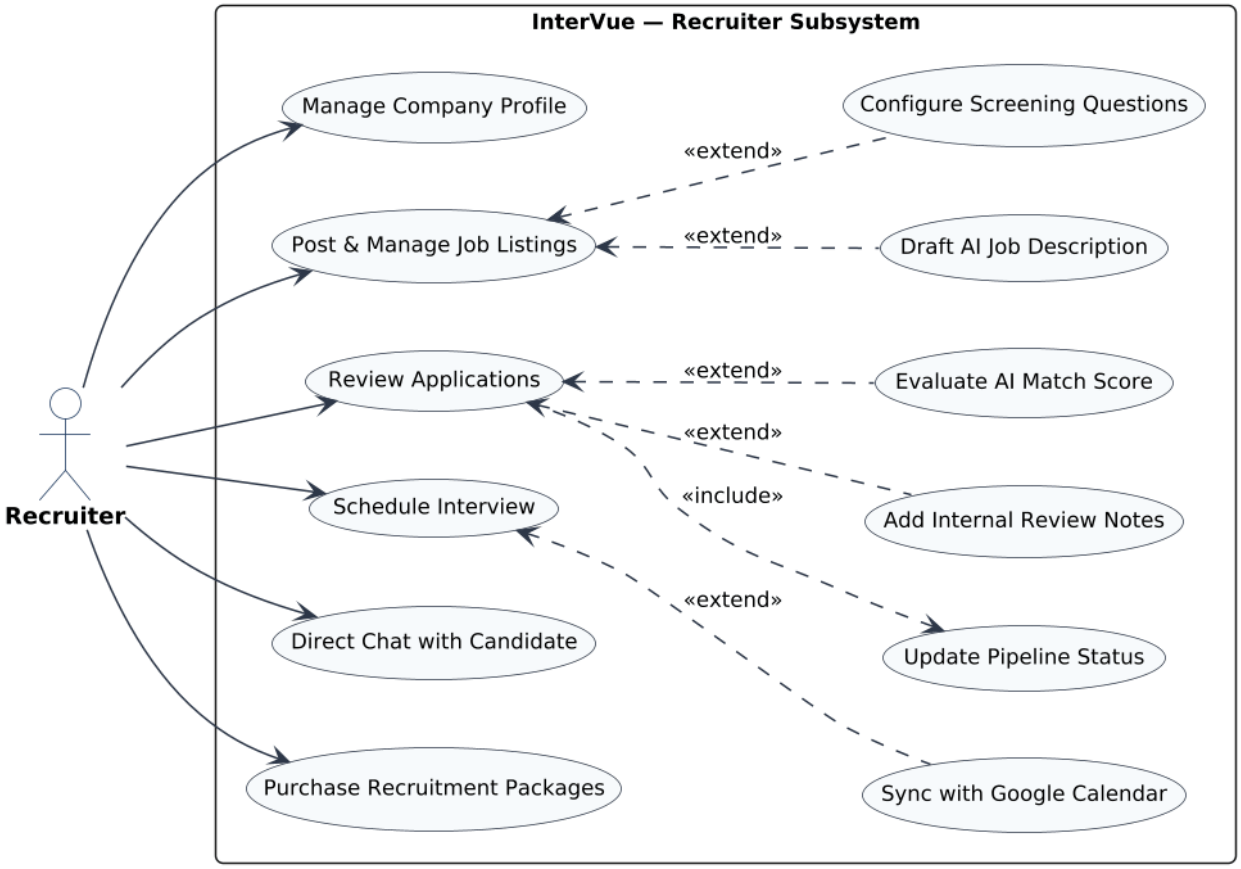


Figure 2.3. Recruiter Subsystem Use Case Diagram

4. Administrator Subsystem Use Case Model

The Administrator Subsystem provides platform operators with governance, moderation, and financial reconciliation tools. The Super Admin actor specializes the Administrator role with exclusive account management authority.

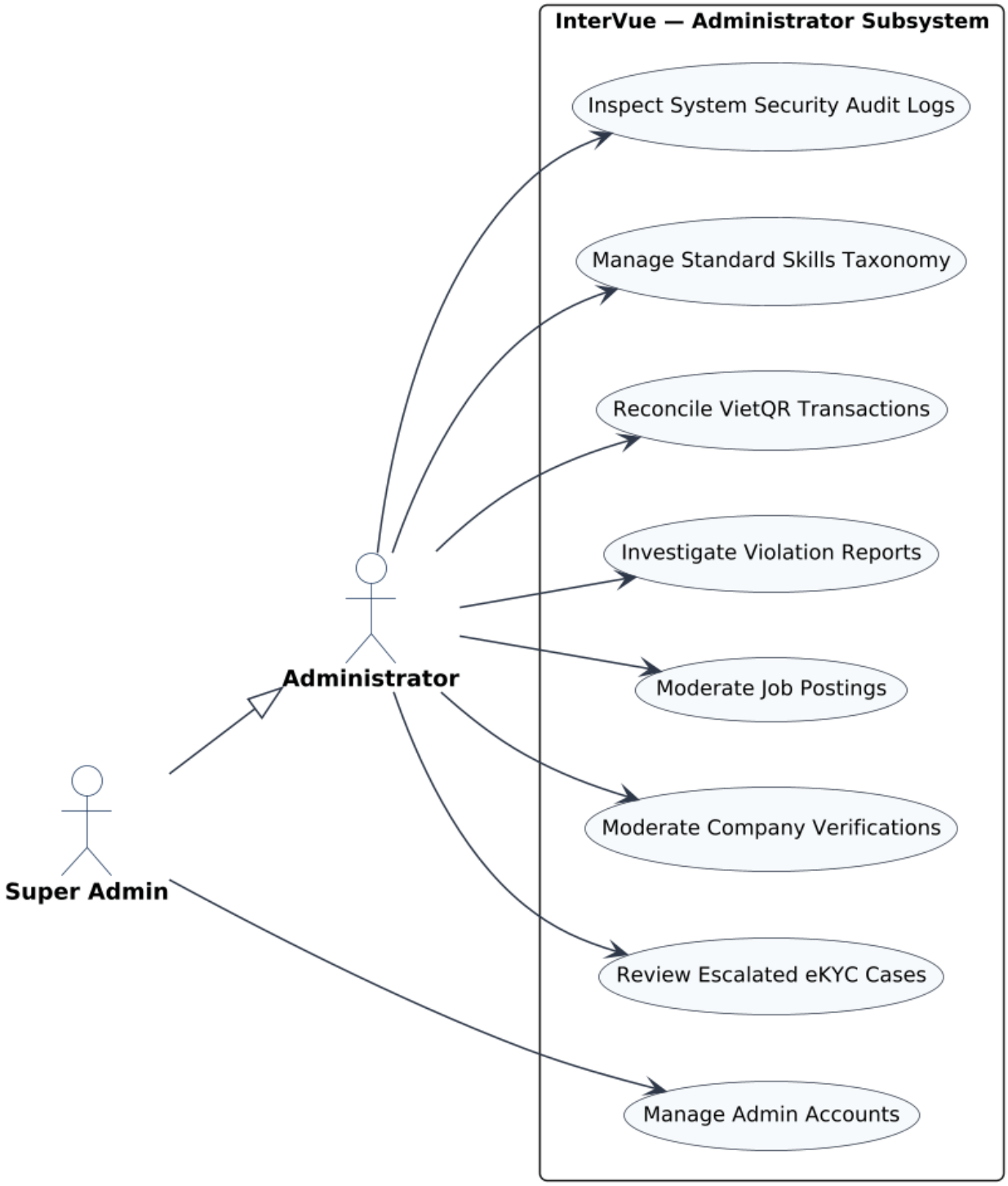


Figure 2.4. Administrator Subsystem Use Case Diagram

III. Sequences Diagrams

1. Guest Subsystem: 1.1. Candidate Registration & Verification

Use Case Name	UC-01: Candidate Registration & Email Verification
Actors	Guest / Candidate
Description	Registers a new candidate account with BCrypt encrypted password and sends verification OTP.
Pre / Post Conditions	Pre: Unauthenticated. Post: Account created (PENDING_VERIFICATION), OTP email dispatched.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Submit registration form (Email, Password, Name, Phone). 2. Client validates field formats & sends POST /api/v1/auth/register/candidate. 3. Auth Service verifies email uniqueness in MySQL database. 4. Hash password with BCrypt (cost 12), save user record & dispatch async OTP email. 5. Return HTTP 201 Created. Client transitions to OTP verification screen.
Exception Flows	- [409 Conflict]: Email already registered -> System returns error, client shows warning. - [400 Bad Request]: Invalid password or email format -> Client halts submission.

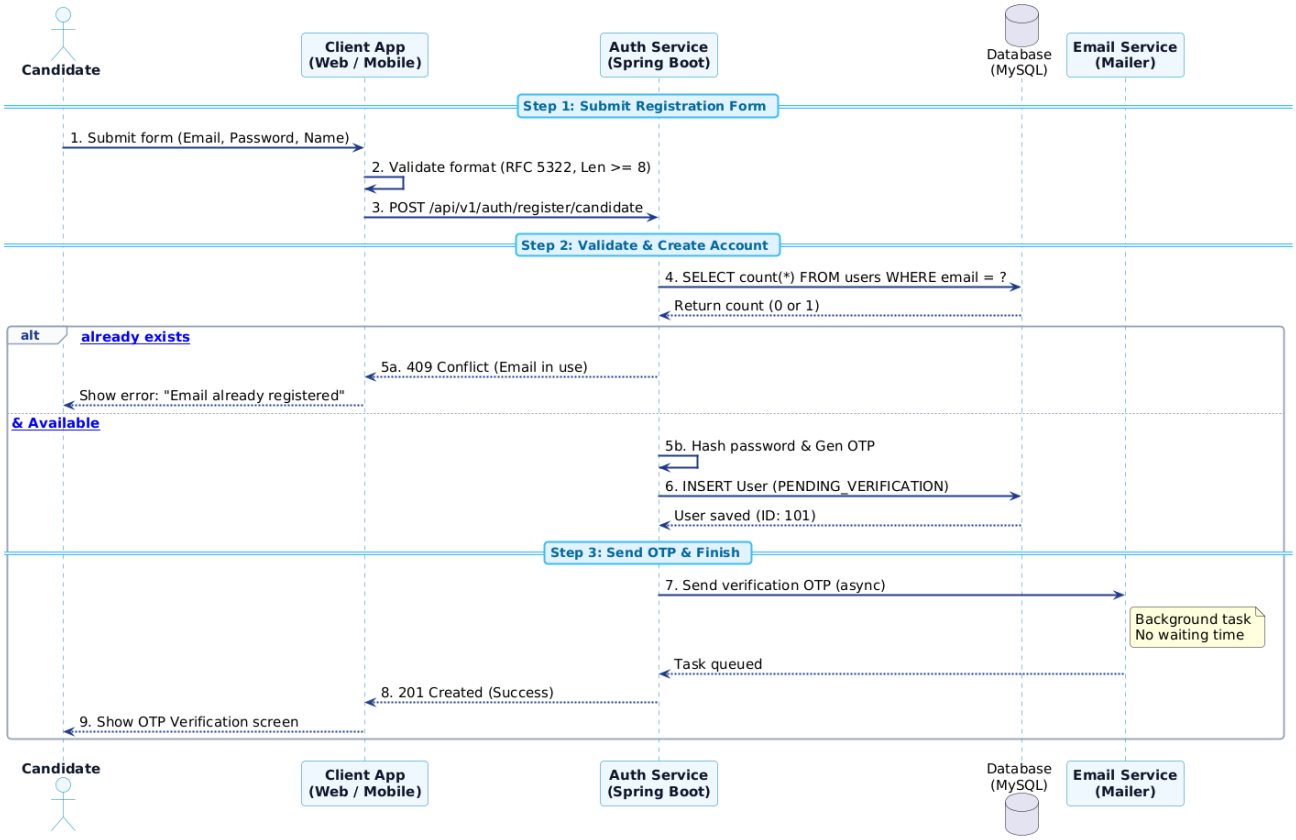


Figure 3.1. Candidate Registration & Account Verification Sequence Diagram

1. Guest Subsystem: 1.2. Recruiter Registration & Company Linkage

Use Case Name	UC-02: Recruiter Registration & Company Linkage
Actors	Guest / Recruiter
Description	Registers an enterprise recruiter account, links to an existing or pending company record.
Pre / Post Conditions	Pre: Unauthenticated. Post: Recruiter account created with role RECRUITER (PENDING_APPROVAL).
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Submit recruiter signup form (Work Email, Company Name, Tax Code, Phone). 2. Client validates tax ID format & sends POST /api/v1/auth/register/recruiter. 3. Auth Service checks email uniqueness and tax code in Company repository. 4. Create pending company dossier, save Recruiter entity & trigger email notification. 5. Return HTTP 201 Created with onboarding instructions.
Exception Flows	- [409 Conflict]: Corporate email already registered -> Return HTTP 409 error. - [422 Unprocessable]: Invalid Tax Code format -> Reject registration attempt.

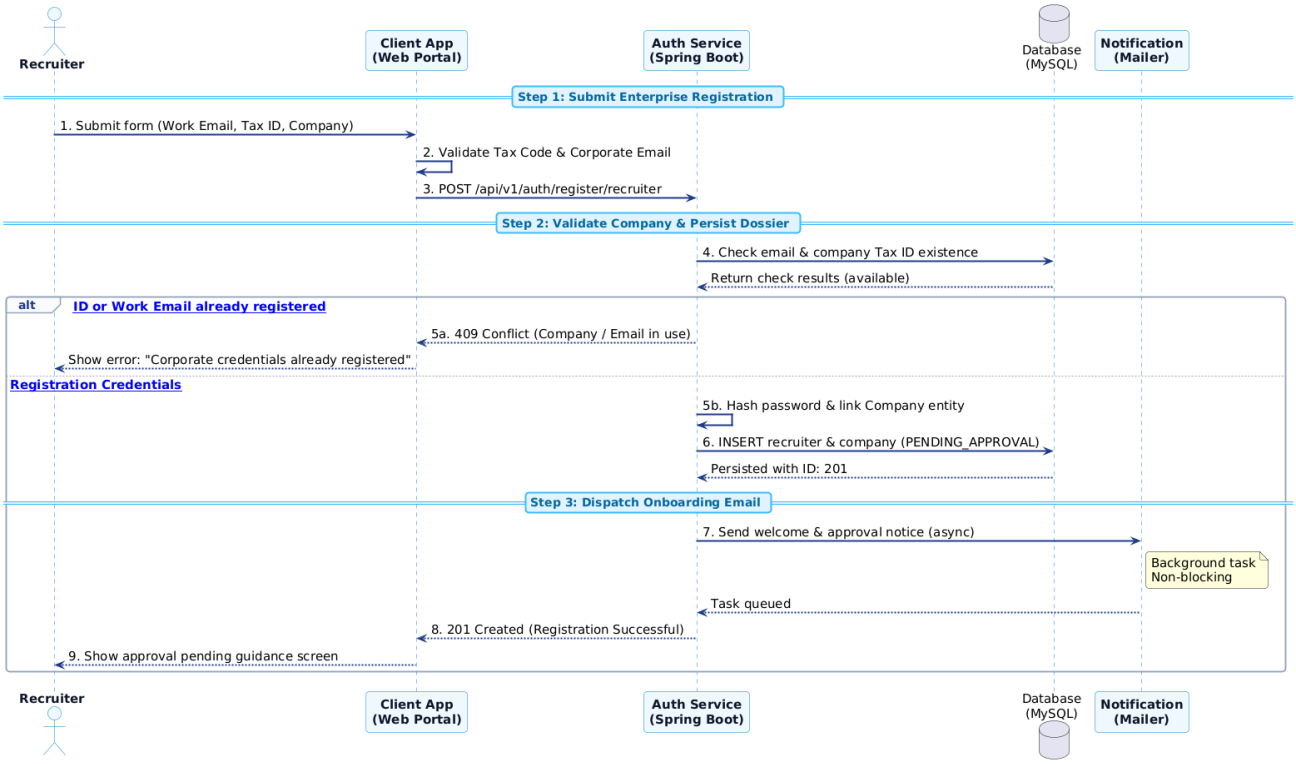


Figure 3.2. Recruiter Registration & Company Linkage Sequence Diagram

1. Guest Subsystem: 1.3. User Authentication & JWT Issuance

Use Case Name	UC-03: User Authentication & JWT Issuance
Actors	Guest / Registered User
Description	Authenticates user credentials, handles 2FA challenges, issues dual JWT access and refresh tokens.
Pre / Post Conditions	Pre: Unauthenticated. Post: Access token (15m) and Refresh token (7d) issued in Redis session.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter email and password, click 'Sign In'. 2. Client dispatches POST /api/v1/auth/login. 3. Auth Service validates BCrypt hash and checks account status (ACTIVE). 4. Issue signed Access Token (RS256, 15m) and store hashed Refresh Token in Redis. 5. Return HTTP 200 OK with JWT token pair and role permissions.
Exception Flows	- [401 Unauthorized]: Bad credentials or wrong password -> Increment failure counter. - [403 Forbidden]: Account suspended or locked -> Return security violation notice.

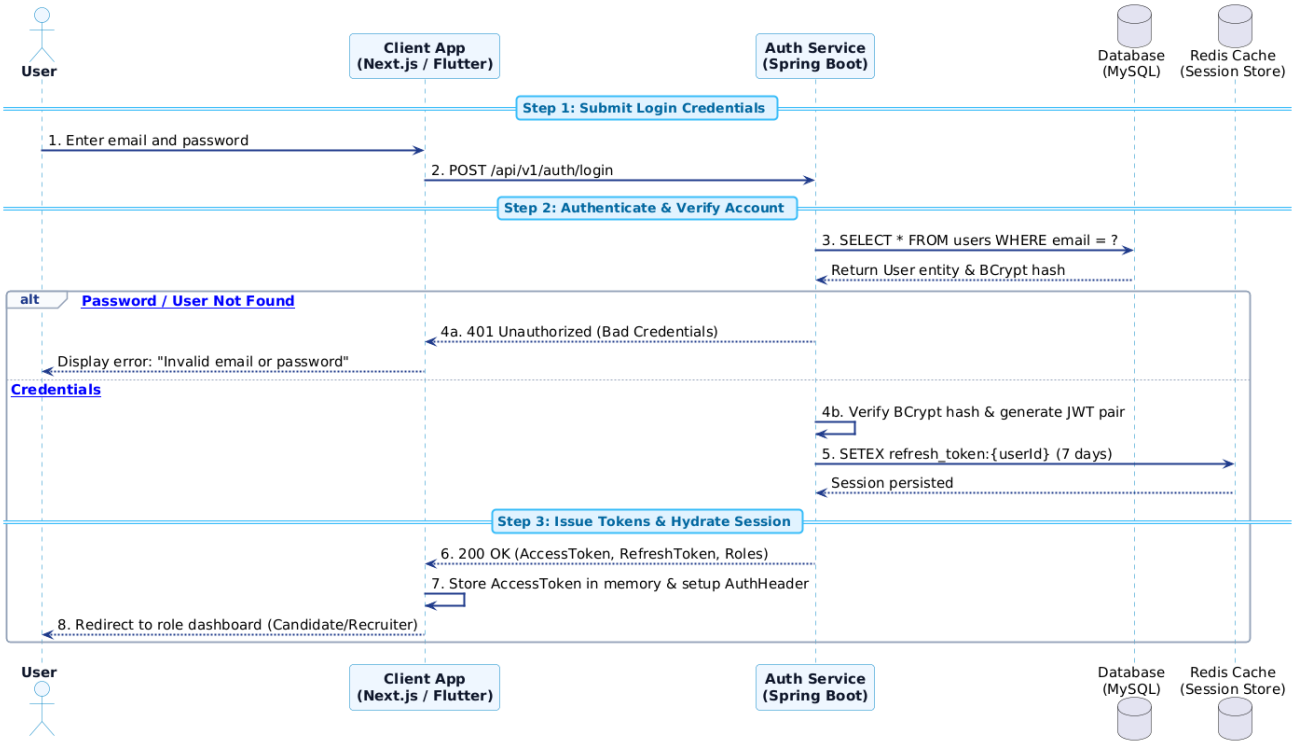


Figure 3.3. User Authentication & JWT Issuance Sequence Diagram

1. Guest Subsystem: 1.4. Public Job Search & Multi-criteria Filtering

Use Case Name	UC-04: Public Job Search & Multi-criteria Filtering
Actors	Guest / Candidate
Description	Searches public active jobs by keyword, salary range, work arrangement, and skill tags.
Pre / Post Conditions	Pre: None. Post: Paginated list of matching job summary cards returned.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter search keyword and select filter criteria (Level, Salary, Location). 2. Client sends GET /api/v1/public/jobs with query parameters. 3. Job Service checks Redis query cache for identical search parameters. 4. Query MySQL jobs with indexed filters, sort by publish date and relevance. 5. Return HTTP 200 OK with paginated list and aggregate facets.
Exception Flows	- [400 Bad Request]: Negative salary range or invalid page size -> Return validation error. - [200 OK Empty]: No jobs match filter -> Return empty array with suggested keywords.

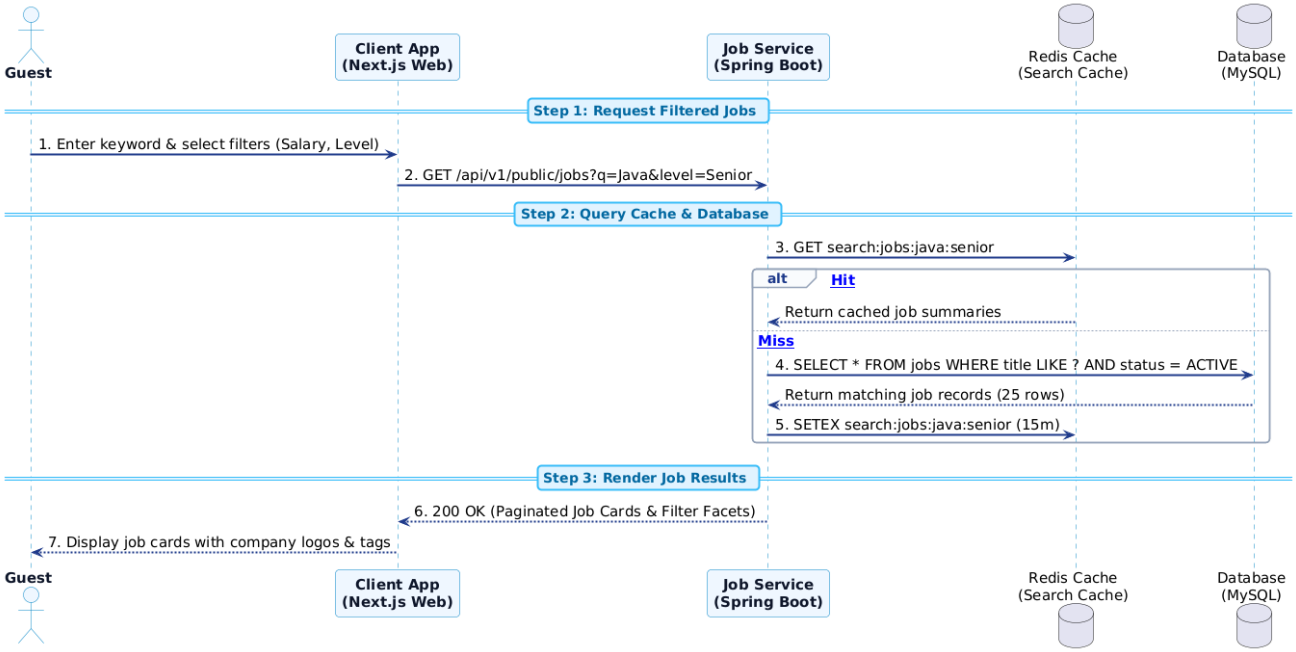


Figure 3.4. Public Job Search & Multi-criteria Filtering Sequence Diagram

2. Candidate Subsystem: 2.1. Password Change & Two-Factor Authentication (2FA)

Use Case Name	UC-05: Password Change & Two-Factor Authentication (2FA)
Actors	Authenticated Candidate
Description	Enables candidates to change account passwords and configure TOTP-based 2FA with QR provisioning.
Pre / Post Conditions	Pre: Candidate authenticated. Post: Password updated, TOTP secret active, recovery codes generated.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter current password and new password, click 'Enable 2FA'. 2. Client dispatches PUT /api/v1/candidate/security/2fa/setup. 3. Security Service verifies current password and generates Base32 TOTP secret. 4. Candidate scans QR on Google Authenticator and enters 6-digit verification code. 5. Service verifies TOTP token, stores encrypted secret & returns 8 recovery codes.
Exception Flows	- [400 Bad Request]: Current password verification fails -> Reject password change. - [422 Unprocessable]: Invalid 6-digit TOTP code during setup -> Reject activation.

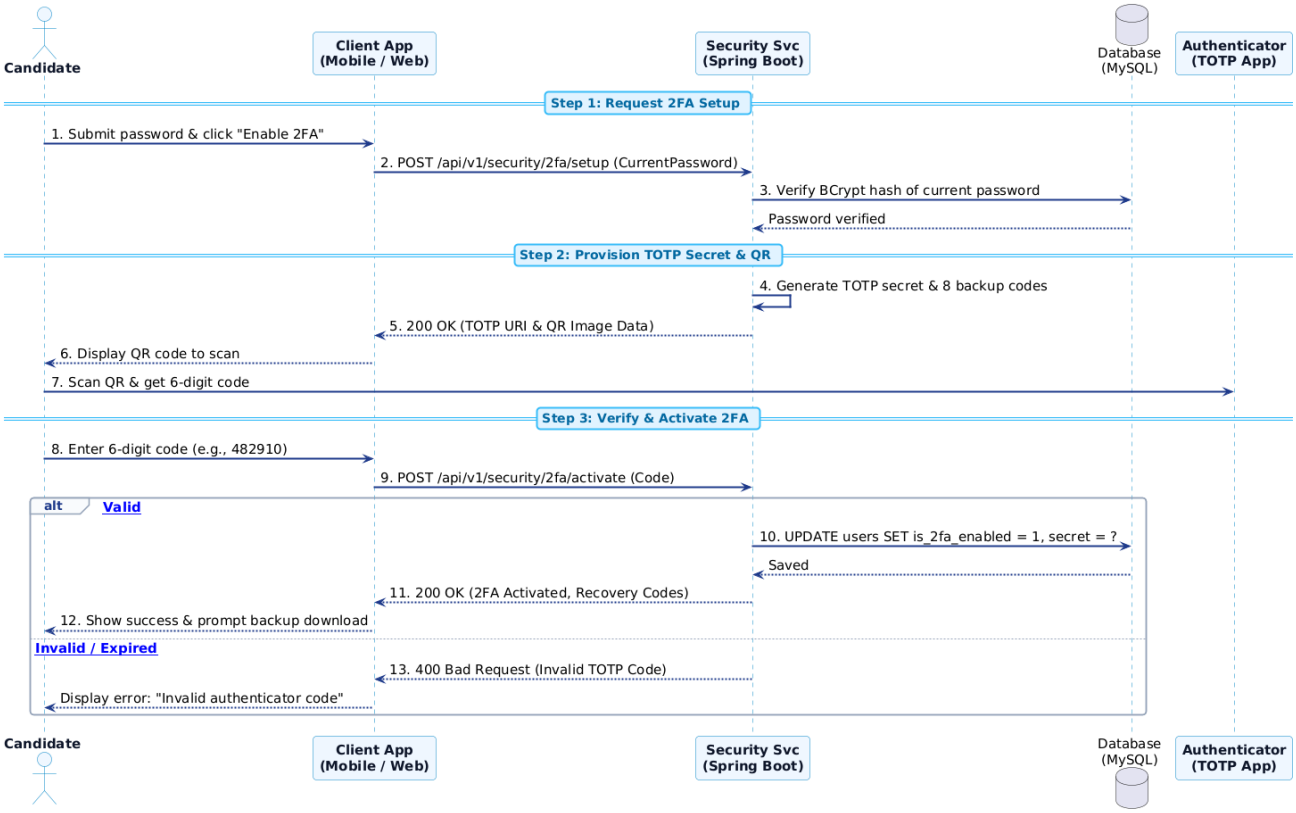


Figure 3.5. Password Change & Two-Factor Authentication Sequence Diagram

2. Candidate Subsystem: 2.2. Identity Verification (eKYC CCCD & Liveness Detection)

Use Case Name	UC-06: Identity Verification (eKYC CCCD & Liveness)
Actors	Authenticated Candidate
Description	Verifies candidate legal identity using National ID OCR and facial liveness detection via FPT.AI.
Pre / Post Conditions	Pre: User logged in. Post: eKYC status updated to VERIFIED (or ESCALATED if score < 80%).
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Upload front/back CCCD photos and perform live face selfie scan. 2. Client dispatches multipart POST /api/v1/candidate/ekyc/verify. 3. eKYC Service routes front/back cards to FPT.AI OCR endpoint for text extraction. 4. eKYC Service invokes FPT.AI FaceMatch comparing portrait on CCCD with live selfie. 5. If FaceMatch >= 80%, update user eKYC status to VERIFIED and unlock job applications.
Exception Flows	- [422 Unprocessable]: CCCD blurred or glare detected -> Request document re-upload. - [409 Conflict]: ID number already bound to another account -> Escalate to manual review.

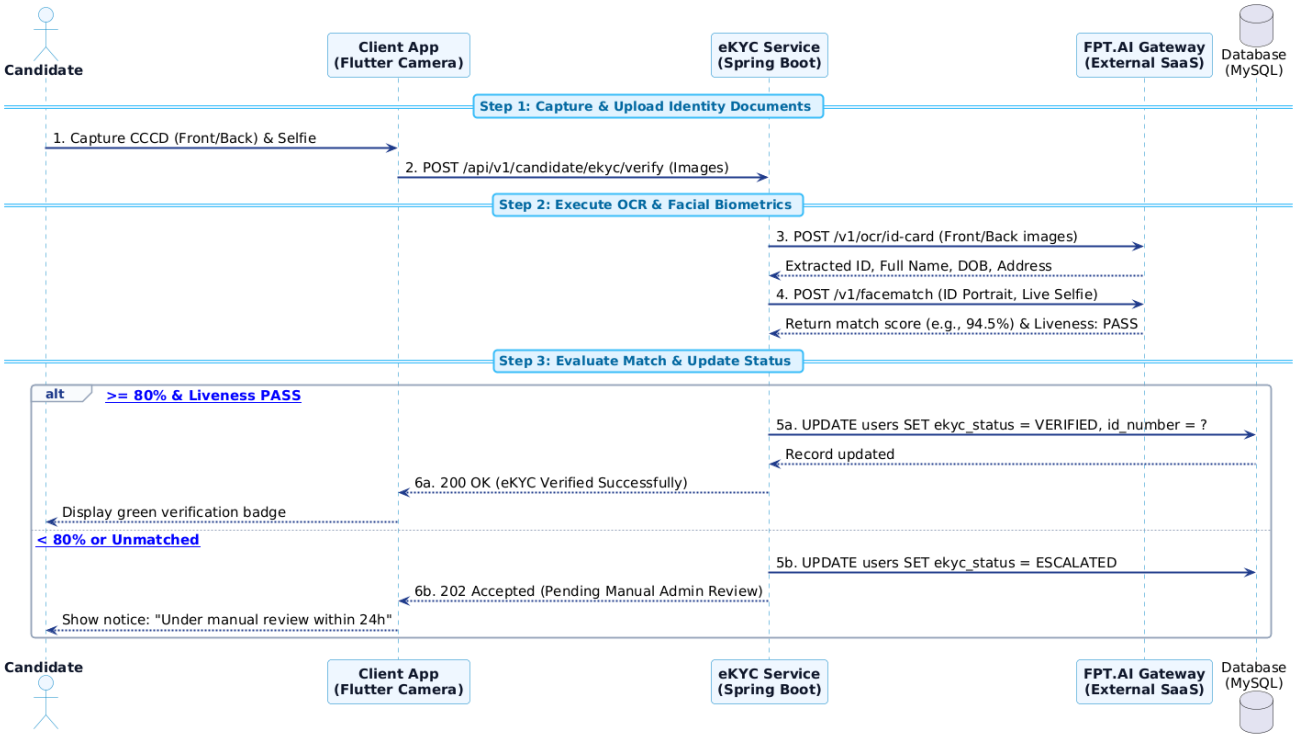


Figure 3.6. Identity Verification (eKYC CCCD & Liveness) Sequence Diagram

2. Candidate Subsystem: 2.3. Candidate Profile Management & Experience Update

Use Case Name	**UC-07: Candidate Profile Management & Experience Update**
Actors	Authenticated Candidate
Description	Maintains candidate technical bio, work experiences, skills, education, and job expectations.
Pre / Post Conditions	Pre: Candidate authenticated. Post: Profile records updated; search indexing cache invalidated.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Edit work experience, education, and technical skills on profile screen. 2. Client dispatches PUT /api/v1/candidate/profile. 3. Candidate Service validates dates and resolves skill taxonomy IDs. 4. Execute batch upsert into candidate_profiles and candidate_skills tables. 5. Evict stale profile cache and return HTTP 200 OK with refreshed profile object.
Exception Flows	- [400 Bad Request]: Start date later than end date -> Return validation error. - [404 Not Found]: Unknown skill tag -> Create custom pending tag or reject.

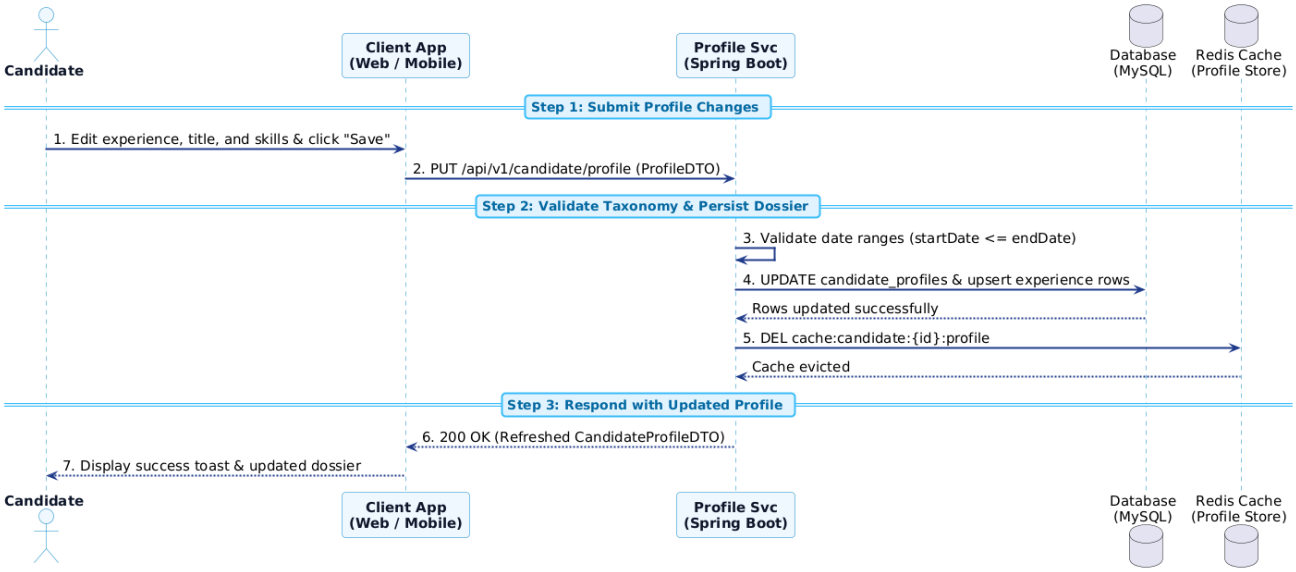


Figure 3.7. Candidate Profile Management & Experience Update Sequence Diagram

2. Candidate Subsystem: 2.4. CV Upload & Default CV Selection

Use Case Name	UC-08: CV Upload & Default CV Selection
Actors	Authenticated Candidate
Description	Uploads PDF resume to secure Cloud Object Storage, performs virus scan, and sets default CV.
Pre / Post Conditions	Pre: Valid PDF (<5MB). Post: CV file persisted in S3/MinIO, record active in database.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Select PDF resume file from local device and click 'Upload CV'. 2. Client validates file size (<=5MB) and type (application/pdf). 3. Upload file via POST /api/v1/candidate/cvs with multipart payload. 4. CV Service uploads binary to Cloud Object Storage and stores metadata in MySQL. 5. Return HTTP 201 Created and set as default CV for rapid job applications.
Exception Flows	- [413 Payload Too Large]: File size exceeds 5MB limit -> Reject upload. - [415 Unsupported Media]: Non-PDF format detected -> Reject with MIME notice.

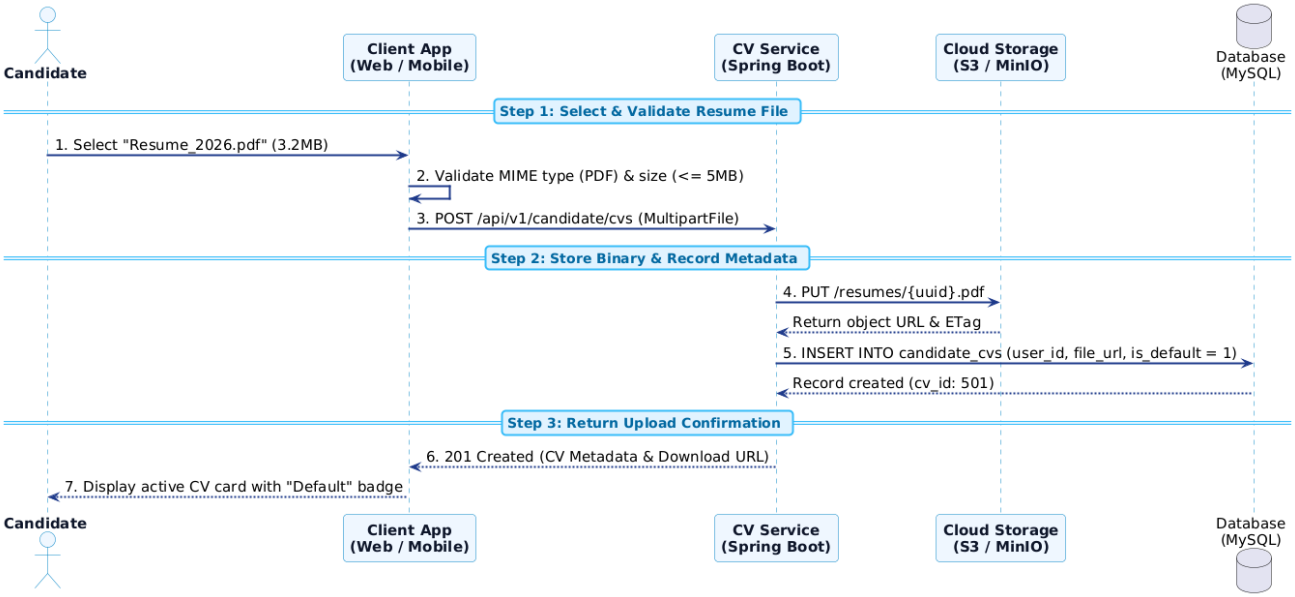


Figure 3.8. CV Upload & Default CV Selection Sequence Diagram

2. Candidate Subsystem: 2.5. Job Application Submission & Screening Q&A

Use Case Name	UC-09: Job Application Submission & Screening Q&A
Actors	Verified Candidate
Description	Submits formal job application with frozen CV snapshot, answers custom screening questions.
Pre / Post Conditions	Pre: Candidate eKYC verified. Post: Application status SUBMITTED, async AI match evaluation queued.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Select CV, answer mandatory screening questions, and click 'Submit Application'. 2. Client dispatches POST /api/v1/candidate/applications. 3. Application Service verifies candidate eKYC status and confirms no duplicate submission. 4. Freeze immutable CV snapshot, persist screening answers, set status SUBMITTED. 5. Emit ApplicationSubmittedEvent to RabbitMQ for async AI scoring and return HTTP 201.
Exception Flows	- [403 Forbidden]: eKYC not completed -> Block application and prompt verification. - [409 Conflict]: Candidate already applied for this job -> Display existing status.

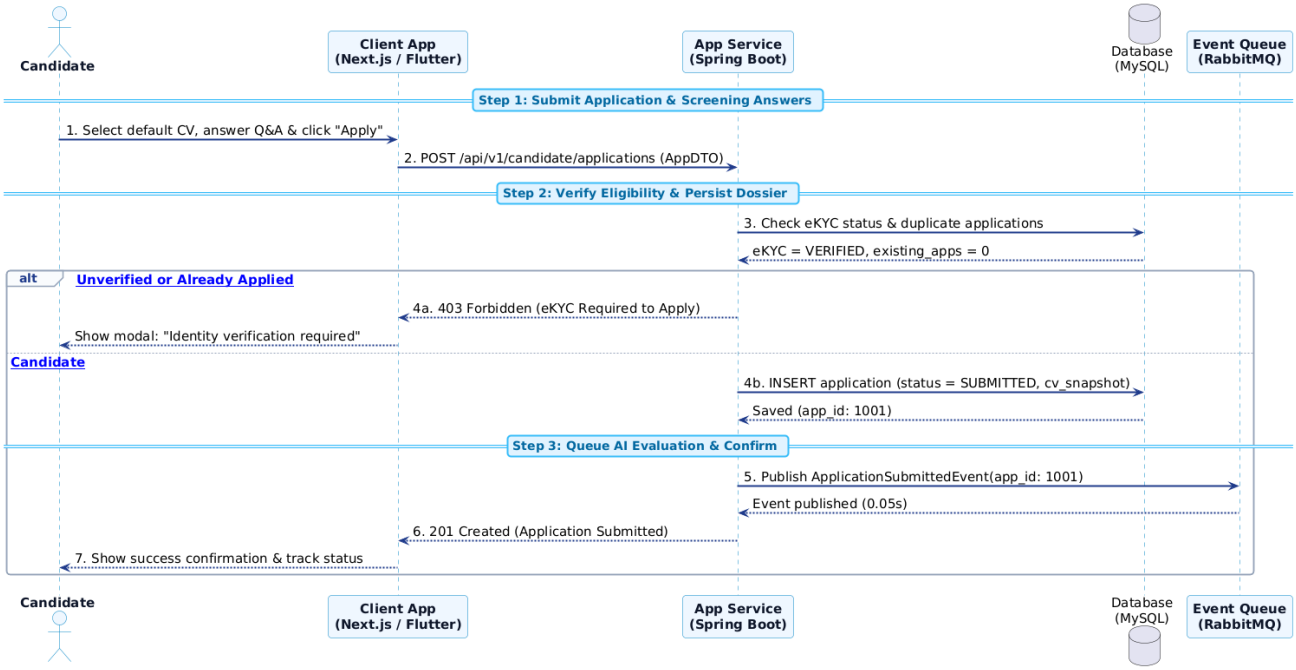


Figure 3.9. Job Application Submission & Screening Q&A Sequence Diagram

2. Candidate Subsystem: 2.6. AI-Powered CV Feedback & Optimization

Use Case Name	UC-10: AI-Powered CV Feedback & Optimization
Actors	Authenticated Candidate
Description	Extracts CV text via OCR, analyzes keyword density, strengths, weaknesses, and ATS compatibility.
Pre / Post Conditions	Pre: Uploaded CV available. Post: Structured AI CV score report and improvement suggestions cached.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Select CV and target job title, click 'Analyze CV with AI'. 2. Client dispatches POST /api/v1/candidate/ai/cv-feedback. 3. AI Service loads CV text and sends structured prompt to Gemini LLM. 4. Gemini LLM returns ATS score, keyword gaps, and bullet point rewrites in JSON format. 5. Save feedback report in MySQL/Redis and return HTTP 200 OK.
Exception Flows	- [422 Unprocessable]: Scanned image without readable text -> Return OCR failure. - [503 Service Unavailable]: Gemini API rate-limited -> Fallback to cached analysis.

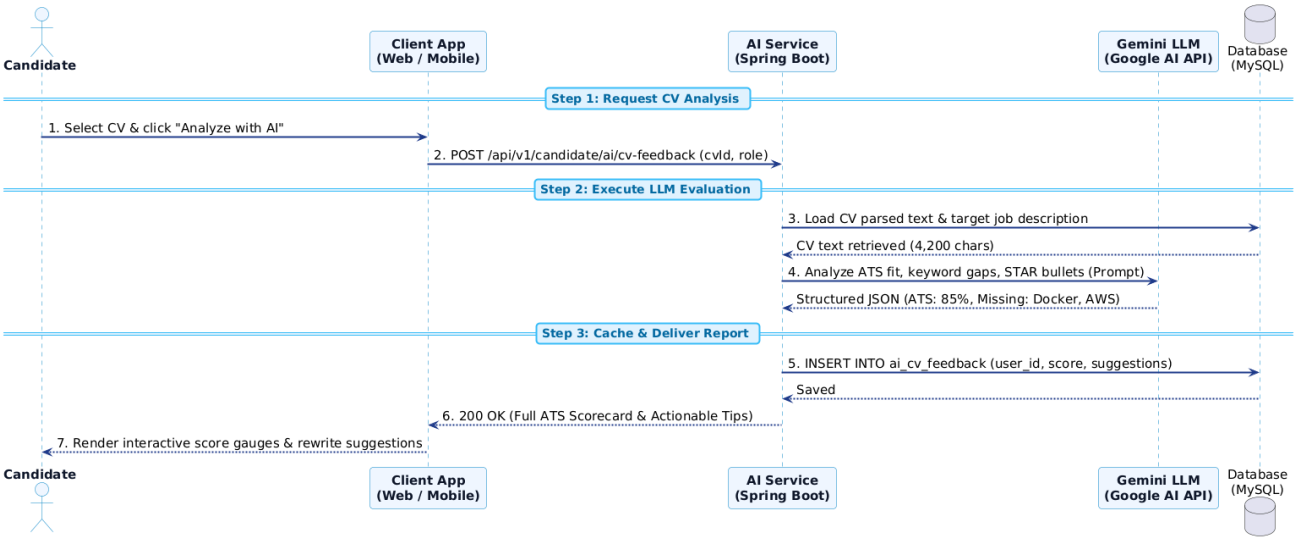


Figure 3.10. AI-Powered CV Feedback & Optimization Sequence Diagram

2. Candidate Subsystem: 2.7. AI Mock Interview Session & Real-time Evaluation

Use Case Name	UC-11: AI Mock Interview Session & Real-time Evaluation
Actors	Authenticated Candidate
Description	Conducts multi-turn conversational mock interview with AI bot, evaluates speech/text answers.
Pre / Post Conditions	Pre: Candidate authenticated. Post: Interview turn recorded, comprehensive feedback scorecard generated.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Select interview domain (e.g. Java Backend), seniority level, and click 'Start Mock Interview'. 2. Client establishes session via POST /api/v1/candidate/mock-interviews/start. 3. AI Service generates dynamic initial interview question tailored to candidate profile. 4. Candidate submits voice/text answer; AI assesses accuracy, depth, and communication tone. 5. After 5 turns, generate summary scorecard (Technical, Behavioral, Problem Solving) with ratings.
Exception Flows	- [400 Bad Request]: Incomplete answer submitted -> Prompt user for more details. - [408 Timeout]: Session idle > 15 minutes -> Automatically finalize and grade session.

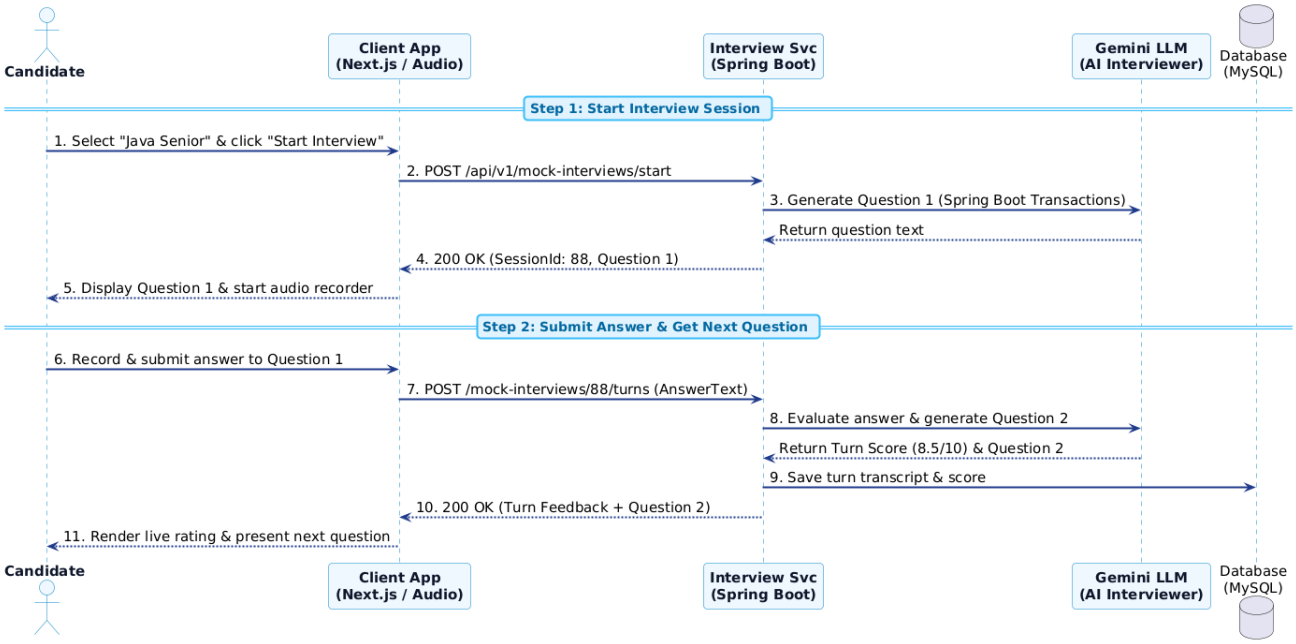


Figure 3.11. AI Mock Interview Session & Real-time Evaluation Sequence Diagram

2. Candidate Subsystem: 2.8. Application Tracking & Interview Schedule Confirmation

Use Case Name	UC-12: Application Tracking & Interview Schedule Confirmation
Actors	Authenticated Candidate
Description	Tracks progress through recruitment stages and confirms/reschedules recruiter interview invitations.
Pre / Post Conditions	Pre: Application exists. Post: Interview status updated to CONFIRMED, Google Meet link dispatched.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. View application details and receive notification of scheduled interview. 2. Candidate reviews proposed time slots (Date, Time, Interviewer, Platform). 3. Client dispatches POST /api/v1/candidate/interviews/{id}/confirm. 4. Interview Service updates status to CONFIRMED and synchronizes calendar invite. 5. Send automated confirmation notification to Recruiter and return HTTP 200 OK.
Exception Flows	- [409 Conflict]: Slot conflict detected -> Request alternate slot selection. - [410 Gone]: Interview slot expired or cancelled by recruiter -> Show update alert.

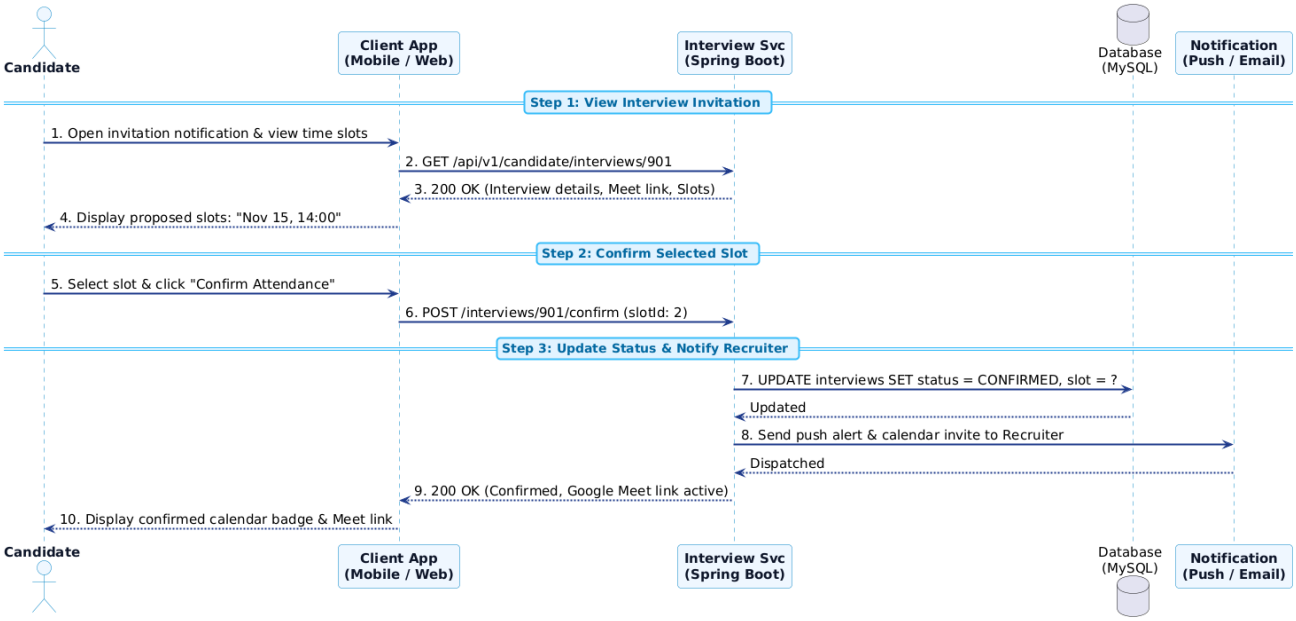


Figure 3.12. Application Tracking & Interview Schedule Confirmation Sequence Diagram

2. Candidate Subsystem: 2.9. Real-time Messaging with Recruiter

Use Case Name	**UC-13: Real-time Messaging with Recruiter**
Actors	Authenticated Candidate
Description	Exchanges instant messages and attachments with recruiter over secure WebSocket connection.
Pre / Post Conditions	Pre: Active application conversation. Post: Message stored in MySQL, delivered via WebSocket/STOMP.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Connect to WebSocket endpoint /ws with JWT authentication token. 2. Candidate types message and dispatches STOMP frame to /app/chat.send. 3. Chat Service validates user membership in conversation room. 4. Persist message entity in MySQL database and update unread counter in Redis. 5. Broadcast message payload to /topic/room/{roomId} for real-time recipient delivery.
Exception Flows	- [401 Unauthorized]: Expired JWT on WebSocket connection -> Disconnect socket. - [403 Forbidden]: Non-member attempts to publish to room -> Discard message.

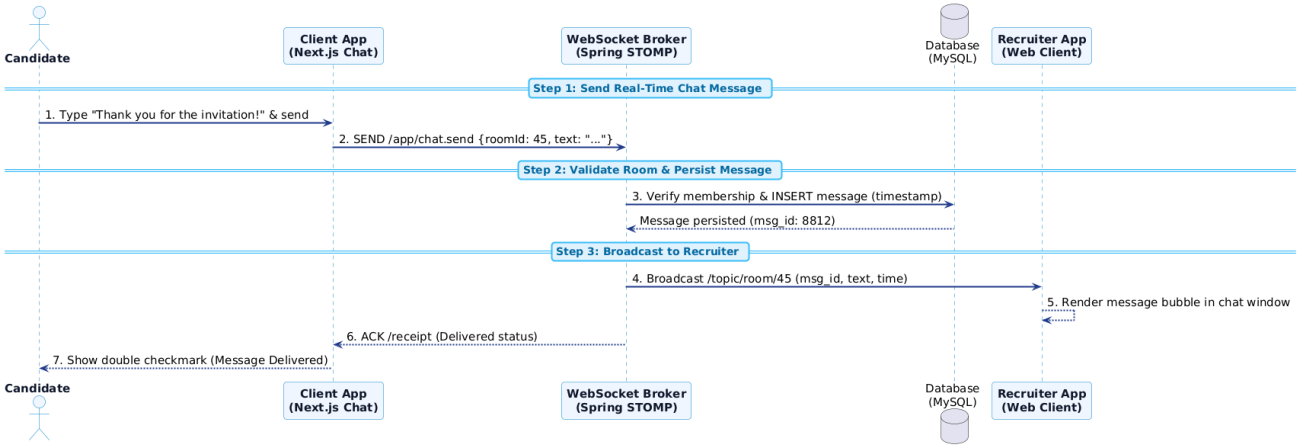


Figure 3.13. Real-time Messaging with Recruiter Sequence Diagram

3. Recruiter Subsystem: 3.1. Company Profile Setup & Business License Verification

Use Case Name	UC-14: Company Profile Setup & Business License Verification
Actors	Authenticated Recruiter
Description	Submits enterprise corporate details, office addresses, and business license PDF for verification.
Pre / Post Conditions	Pre: Recruiter authenticated. Post: Company profile updated, status PENDING_VERIFICATION.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter company bio, industry, website, addresses, and upload business license PDF. 2. Client sends multipart PUT /api/v1/recruiter/company/profile. 3. Company Service uploads license document to Cloud Storage and updates company record. 4. System creates an administrative review task in the moderation queue. 5. Return HTTP 200 OK with company profile and 'Pending Verification' badge.
Exception Flows	- [413 Payload Too Large]: License file > 10MB -> Return file size limit error. - [400 Bad Request]: Missing required tax registration number -> Reject update.

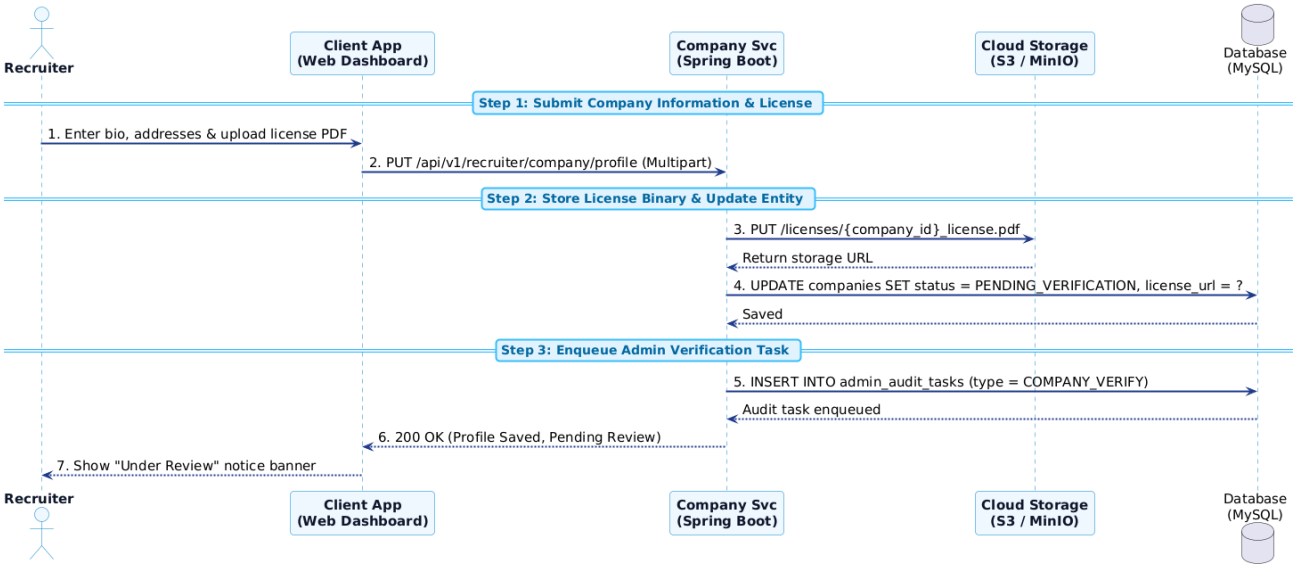


Figure 3.14. Company Profile Setup & Business License Verification Sequence Diagram

3. Recruiter Subsystem: 3.2. Job Posting Creation & Screening Questionnaire Builder

Use Case Name	UC-15: Job Posting Creation & Screening Questionnaire Builder
Actors	Verified Recruiter
Description	Creates technical job posting, defines salary bands, tags skills, and builds screening questions.
Pre / Post Conditions	Pre: Recruiter company verified. Post: Job entity created with status ACTIVE, indexed for search.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter job title, description, skills, salary range, and add 3 custom screening questions. 2. Client dispatches POST /api/v1/recruiter/jobs. 3. Job Service validates recruiter posting quota and verifies skill taxonomy IDs. 4. Insert job record, associate skills in job_skills, and insert screening_questions. 5. Invalidate search cache and return HTTP 201 Created with generated job slug.
Exception Flows	- [403 Forbidden]: Company not verified or job quota exhausted -> Prompt package upgrade. - [400 Bad Request]: Min salary exceeds max salary -> Reject form submission.

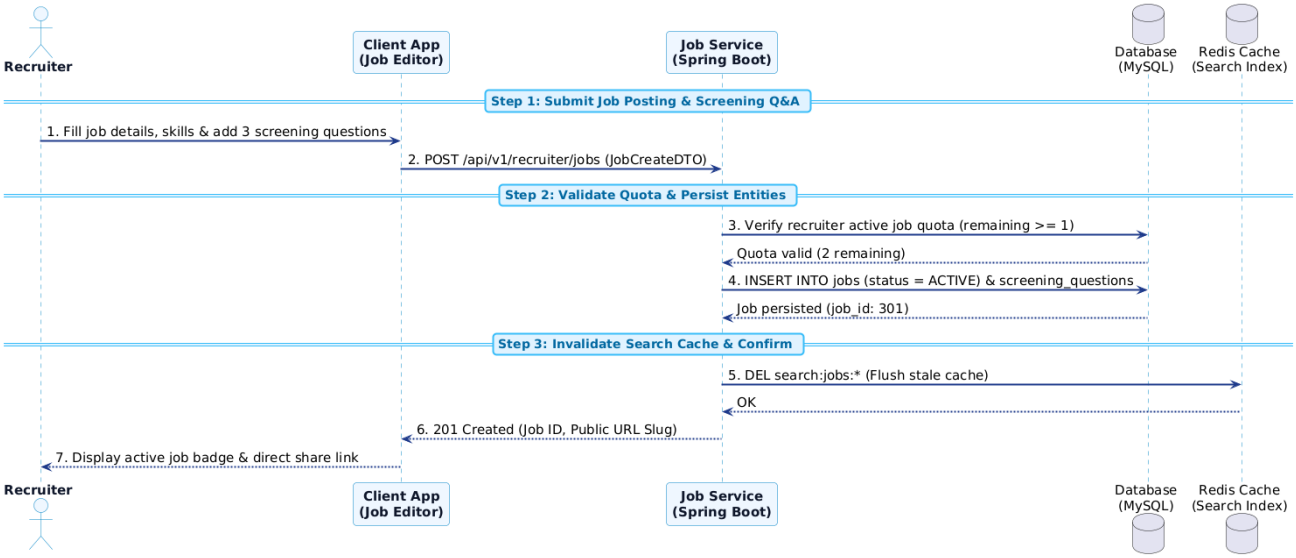


Figure 3.15. Job Posting Creation & Screening Questionnaire Builder Sequence Diagram

3. Recruiter Subsystem: 3.3. Job Posting Lifecycle Management (Edit, Close, Reopen)

Use Case Name	**UC-16: Job Posting Lifecycle Management**
Actors	Authenticated Recruiter
Description	Manages operational status of job openings: editing descriptions, closing, or reopening postings.
Pre / Post Conditions	Pre: Recruiter owns job. Post: Job status toggled (ACTIVE/CLOSED), candidate listings updated.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Select active job posting and click 'Close Job Posting'. 2. Client dispatches PATCH /api/v1/recruiter/jobs/{id}/status {status: 'CLOSED'}. 3. Job Service verifies recruiter ownership of the targeted job entity. 4. Update job status to CLOSED and timestamp closed_at in database. 5. Evict public job cache and return HTTP 200 OK.
Exception Flows	- [403 Forbidden]: User does not belong to job's parent company -> Access denied. - [404 Not Found]: Invalid job ID provided -> Return resource missing.

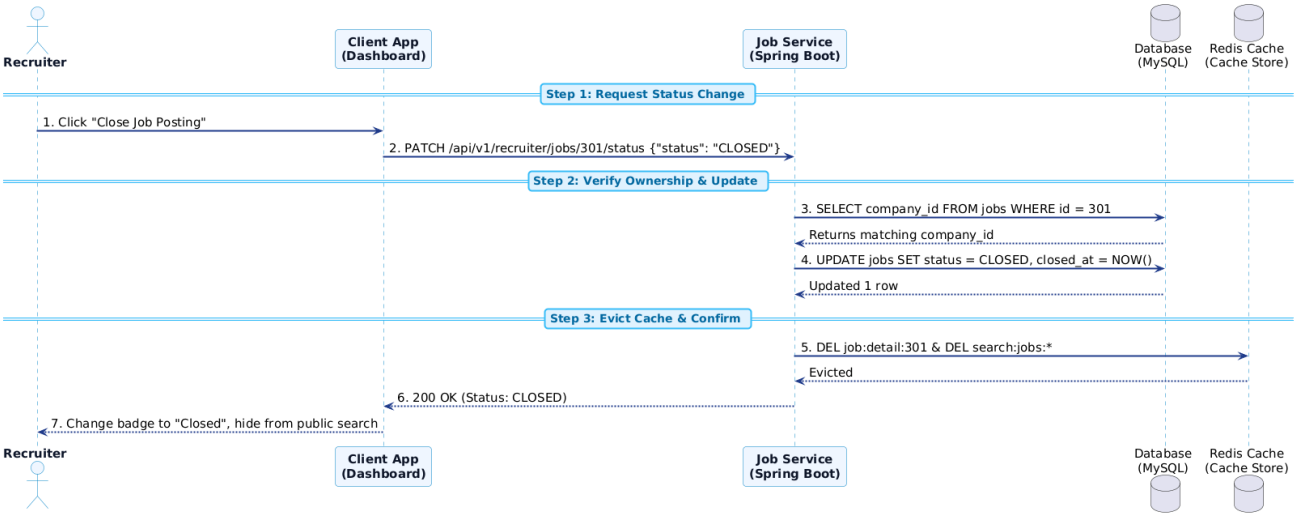


Figure 3.16. Job Posting Lifecycle Management Sequence Diagram

3. Recruiter Subsystem: 3.4. Candidate Pipeline Management (Kanban Stage Transitions)

Use Case Name	UC-17: Candidate Pipeline Management
Actors	Authenticated Recruiter
Description	Moves applications across recruitment stages (REVIEWING, SHORTLISTED, INTERVIEWING, OFFERED, REJECTED).
Pre / Post Conditions	Pre: Application active. Post: Stage updated, audit log saved, automated candidate notice sent.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Drag and drop candidate card from 'Applied' to 'Shortlisted' on Kanban board. 2. Client dispatches PATCH /api/v1/recruiter/applications/{id}/stage {stage: 'SHORTLISTED'}. 3. Application Service validates stage transition rules and updates application record. 4. Insert entry into application_stage_history for audit trail. 5. Trigger candidate email notification asynchronously and return HTTP 200 OK.
Exception Flows	- [400 Bad Request]: Illegal stage transition (e.g., REJECTED -> OFFERED) -> Reject move. - [404 Not Found]: Application does not belong to recruiter job -> Return 404.

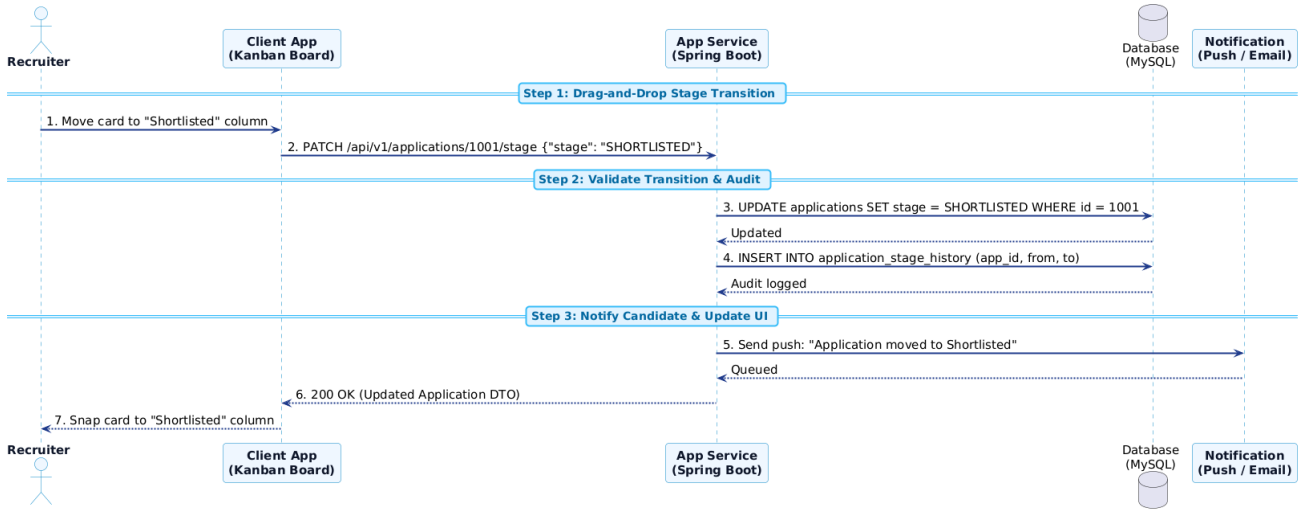


Figure 3.17. Candidate Pipeline Management Sequence Diagram

3. Recruiter Subsystem: 3.5. AI Match Fit Evaluation & Candidate Scorecard Review

Use Case Name	UC-18: AI Match Fit Evaluation & Scorecard Review
Actors	Authenticated Recruiter
Description	Retrieves multi-dimensional AI scoring matrix (Skills, Experience, Screening Q&A, Overall Fit).
Pre / Post Conditions	Pre: Application submitted. Post: AI match scorecard returned with breakdown radar chart.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Click 'View AI Match Scorecard' on candidate dossier in pipeline. 2. Client dispatches GET /api/v1/recruiter/applications/{id}/ai-evaluation. 3. AI Evaluation Service queries ai_match_evaluations table by application_id. 4. If evaluation exists, format breakdown radar metrics and executive summary. 5. Return HTTP 200 OK with match percentage, strength tags, and risk factors.
Exception Flows	- [202 Accepted]: AI scoring still executing in background -> Return 'Processing' status. - [404 Not Found]: Invalid application ID provided -> Return error.

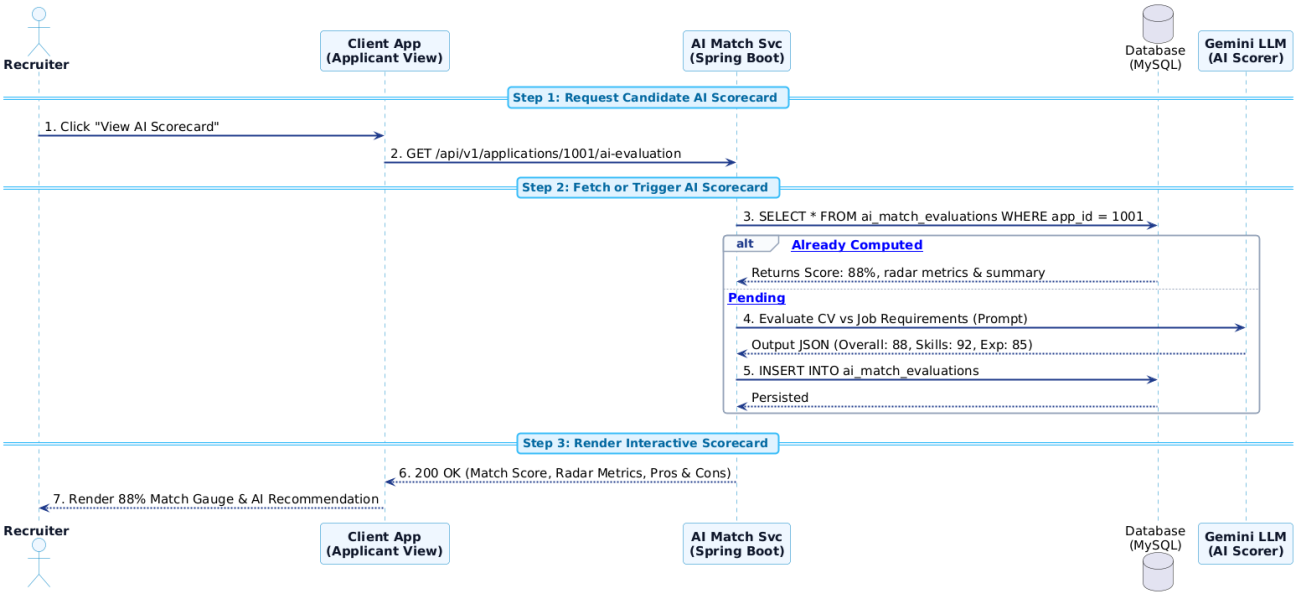


Figure 3.18. AI Match Fit Evaluation & Scorecard Review Sequence Diagram

3. Recruiter Subsystem: 3.6. Interview Scheduling & Invitation Dispatch

Use Case Name	**UC-19: Interview Scheduling & Invitation Dispatch**
Actors	Authenticated Recruiter
Description	Proposes interview dates/times, creates Google Meet / Zoom meeting room, and dispatches invite.
Pre / Post Conditions	Pre: Application shortlisted. Post: Interview entity created (SCHEDULED), candidate notified.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter interview title, format (Online Meet), select 2 candidate time slots, and submit. 2. Client dispatches POST /api/v1/recruiter/interviews. 3. Interview Service validates recruiter calendar and generates secure Google Meet link. 4. Insert interview record with status SCHEDULED and store proposed slot options. 5. Dispatch email invitation with accept/decline action buttons and return HTTP 201
Exception Flows	- [400 Bad Request]: Proposed time slots in the past -> Reject invitation creation. - [409 Conflict]: Overlapping interview with another candidate -> Warn recruiter.

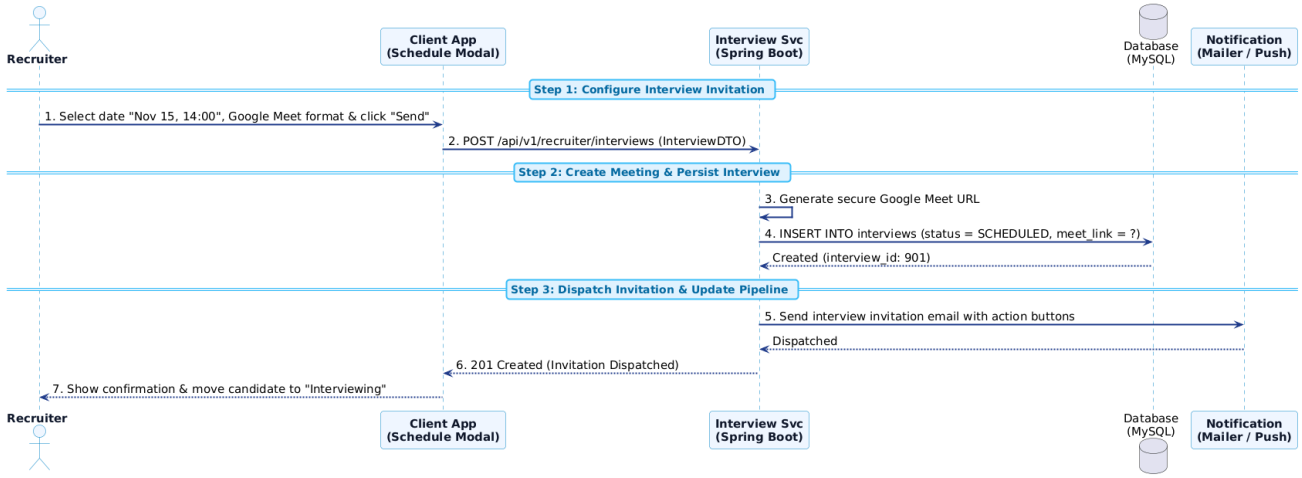


Figure 3.19. Interview Scheduling & Invitation Dispatch Sequence Diagram

3. Recruiter Subsystem: 3.7. Service Package Purchase via SePay Dynamic VietQR

Use Case Name	UC-20: Service Package Purchase via SePay Dynamic VietQR
Actors	Authenticated Recruiter
Description	Generates dynamic VietQR code for recruitment package purchases, processes automated bank webhook.
Pre / Post Conditions	Pre: Order initiated. Post: Webhook verified, order COMPLETED, recruiter quota credited.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Select package (e.g., Enterprise 10 Job Slots) and click 'Pay via VietQR'. 2. Client dispatches POST /api/v1/recruiter/payments/create-order. 3. Payment Service creates order record (PENDING) and generates dynamic VietQR image with code. 4. Recruiter scans VietQR and transfers via mobile banking app (Napas 247). 5. SePay sends webhook; verify HMAC signature, credit quota, and return HTTP 200.
Exception Flows	- [400 Bad Request]: Webhook HMAC signature invalid -> Reject transaction. - [409 Conflict]: Duplicate webhook idempotency key -> Acknowledge without double crediting.

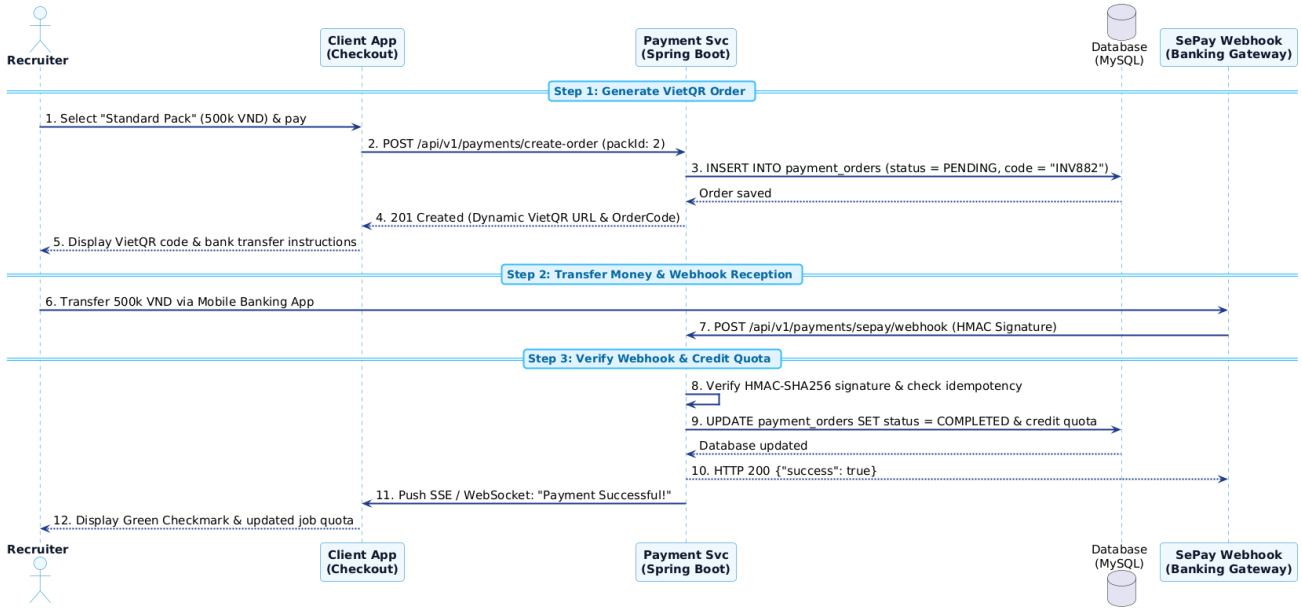


Figure 3.20. Service Package Purchase via SePay Dynamic VietQR Sequence Diagram

3. Recruiter Subsystem: 3.8. Candidate Search & Direct Outreach

Use Case Name	**UC-21: Candidate Search & Direct Outreach**
Actors	Verified Recruiter
Description	Searches talent pool using skill tags and experience filters, consumes quota to send direct job invites.
Pre / Post Conditions	Pre: Recruiter has contact quota. Post: Candidate notified of direct interview interest.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Search candidate database by skill (React, Go) and years of experience. 2. Client dispatches GET /api/v1/recruiter/talents/search. 3. Talent Service queries verified candidate profiles matching criteria. 4. Recruiter selects candidate profile and clicks 'Send Direct Opportunity Invite'. 5. Deduct 1 outreach credit, save direct invitation, and notify candidate via email/app.
Exception Flows	- [403 Forbidden]: Candidate opted out of recruiter discovery -> Hide contact button. - [402 Payment Required]: Outreach quota exhausted -> Prompt package purchase.

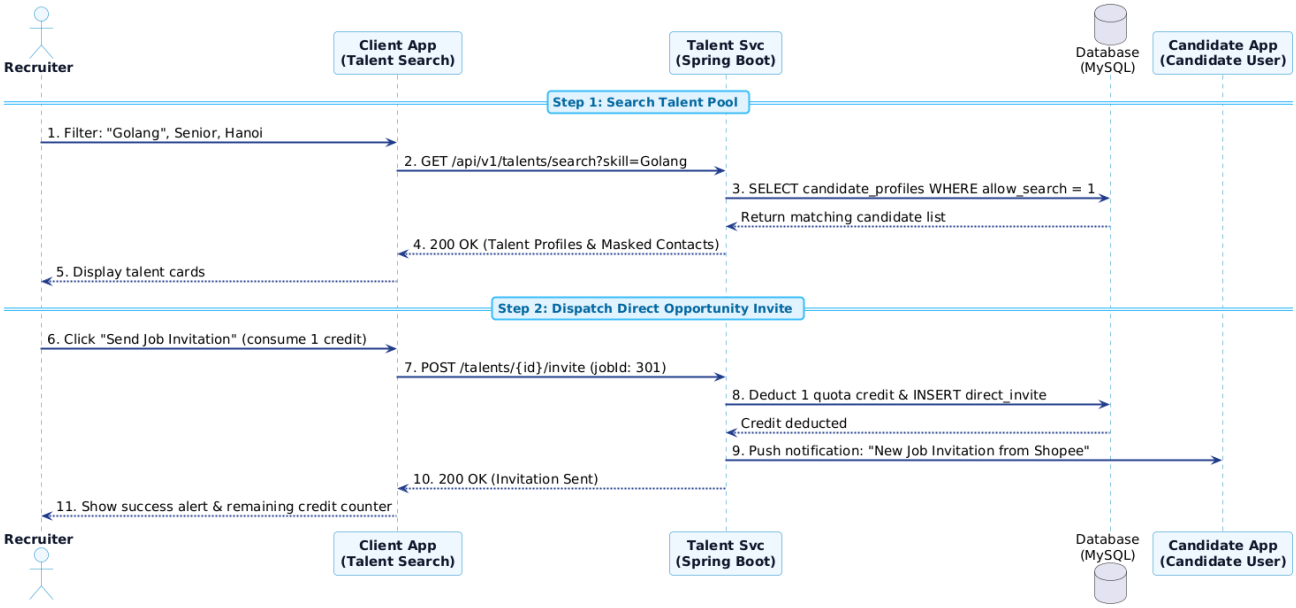


Figure 3.21. Candidate Search & Direct Outreach Sequence Diagram

3. Recruiter Subsystem: 3.9. Direct In-App Chat & Notification with Candidates

Use Case Name	UC-22: Direct In-App Chat & Notification
Actors	Authenticated Recruiter
Description	Engages in real-time conversation with shortlisted applicants regarding interview logistics.
Pre / Post Conditions	Pre: Mutual application connection. Post: Message delivered via WebSocket, stored in history.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Open chat conversation with candidate from application pipeline card. 2. Client connects to WebSocket and sends STOMP frame to /app/chat.send. 3. Chat Service verifies recruiter authorization for this application thread. 4. Insert message record into MySQL and broadcast to /topic/room/{roomId}. 5. If candidate offline, trigger asynchronous push notification to mobile device.
Exception Flows	- [403 Forbidden]: Conversation closed or candidate rejected -> Read-only archive. - [413 Payload Too Large]: Chat attachment exceeds 10MB -> Reject attachment.

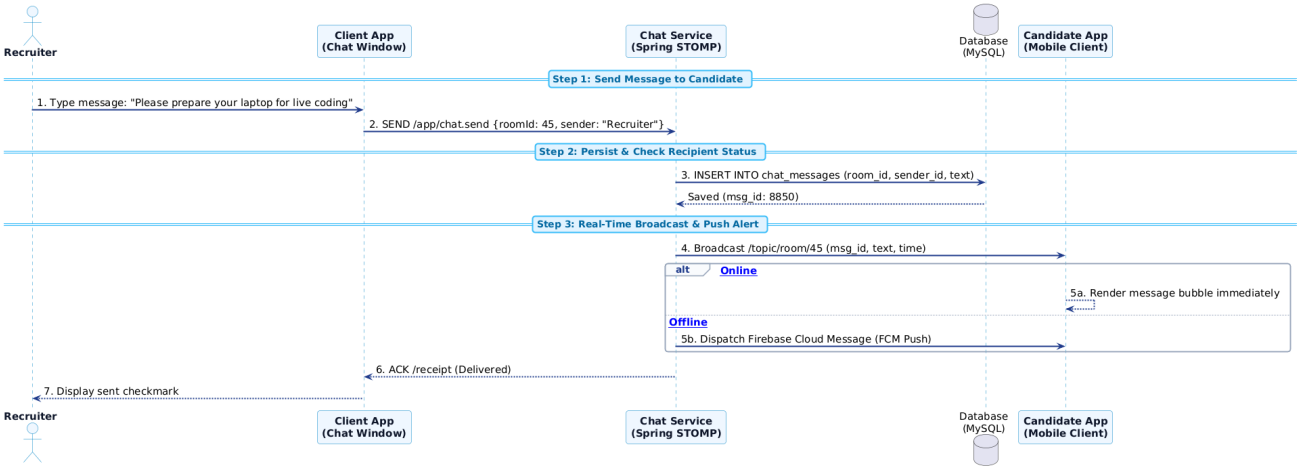


Figure 3.22. Direct In-App Chat & Notification Sequence Diagram

4. Administrator Subsystem: 4.1. User Account Governance (Lock, Unlock, Role Assignment)

Use Case Name	UC-23: User Account Governance
Actors	System Administrator
Description	Audits platform user accounts, suspends fraudulent users, resets compromised accounts, and modifies roles.
Pre / Post Conditions	Pre: Admin authenticated with ROLE_ADMIN. Post: User status updated, active sessions invalidated.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Search user in governance console by email or ID, click 'Lock Account'. 2. Client dispatches PATCH /api/v1/admin/users/{id}/status {status: 'LOCKED', reason: 'Abuse'}. 3. Admin Service verifies caller has ROLE_ADMIN and target is not super admin. 4. Update user status in database and revoke all active refresh tokens in Redis. 5. Log administrative action in audit_logs table and return HTTP 200 OK.
Exception Flows	- [403 Forbidden]: Non-super admin attempts to lock super admin account -> Reject. - [404 Not Found]: User ID does not exist -> Return error.

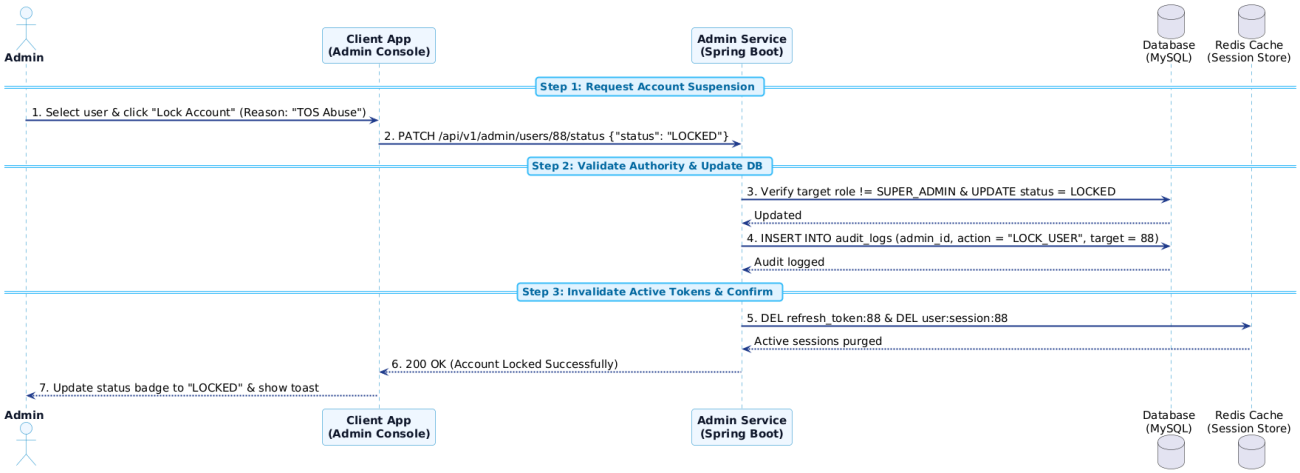


Figure 3.23. User Account Governance Sequence Diagram

4. Administrator Subsystem: 4.2. Manual eKYC Verification Escalation & Audit Review

Use Case Name	**UC-24: Manual eKYC Verification Escalation & Audit Review**
Actors	System Administrator
Description	Reviews flagged eKYC submissions where automated FaceMatch scores fell between 60% and 79%.
Pre / Post Conditions	Pre: Submission escalated. Post: eKYC status decided (VERIFIED or REJECTED) with audit reason.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Open eKYC escalation audit queue, review side-by-side CCCD photo and live selfie. 2. Administrator clicks 'Approve eKYC' or 'Reject eKYC' with rejection note. 3. Client dispatches POST /api/v1/admin/ekyc/{id}/review {decision: 'APPROVE'}. 4. eKYC Service updates user record to VERIFIED and sets reviewer_admin_id. 5. Dispatch email notification to candidate and return HTTP 200 OK.
Exception Flows	- [409 Conflict]: Submission already audited by another concurrent admin -> Return warning. - [400 Bad Request]: Missing rejection reason when rejecting -> Require explanation.

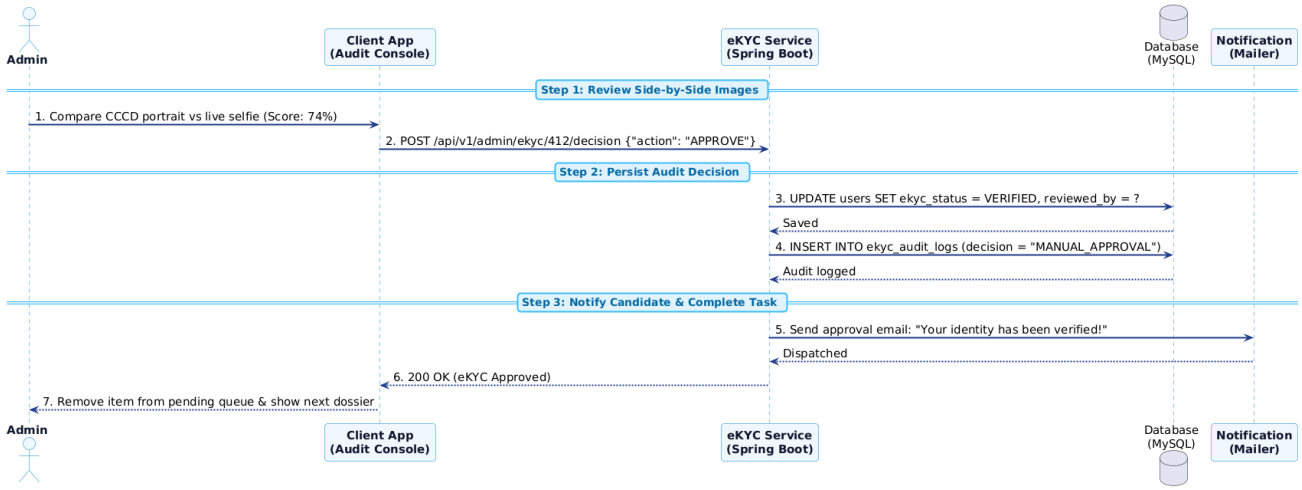


Figure 3.24. Manual eKYC Verification Escalation & Audit Review Sequence Diagram

4. Administrator Subsystem: 4.3. Recruiter Company Verification & License Approval

Use Case Name	UC-25: Recruiter Company Verification & License Approval
Actors	System Administrator
Description	Validates company business licenses against national tax registry, approves or rejects recruiter profiles.
Pre / Post Conditions	Pre: Company submitted license. Post: Company status updated to VERIFIED, recruiter job posting unlocked.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Inspect business license PDF and cross-check company tax code with government portal. 2. Administrator clicks 'Approve Company Verification'. 3. Client dispatches POST /api/v1/admin/companies/{id}/verify {decision: 'APPROVE'}. 4. Company Service updates company status to VERIFIED and grants active posting badge. 5. Dispatch corporate congratulatory email to recruiter and return HTTP 200 OK.
Exception Flows	- [400 Bad Request]: Rejection without specifying invalid license criteria -> Require note. - [404 Not Found]: Company profile not found -> Return error.

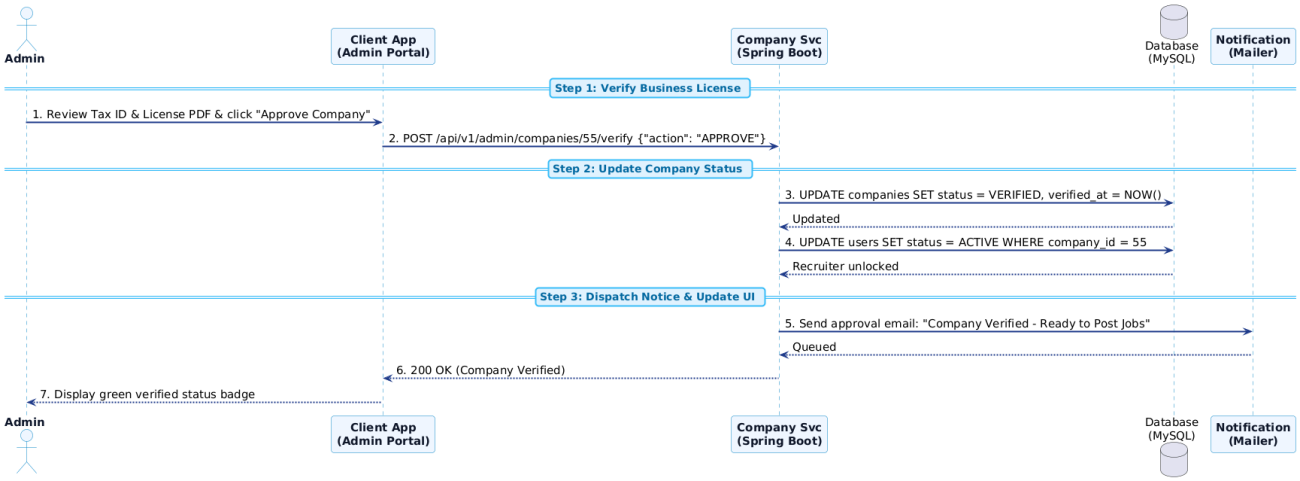


Figure 3.25. Recruiter Company Verification & License Approval Sequence Diagram

4. Administrator Subsystem: 4.4. Job Posting Moderation & Content Compliance Check

Use Case Name	**UC-26: Job Posting Moderation & Content Compliance Check**
Actors	System Administrator
Description	Reviews job postings for policy violations, misleading descriptions, discriminatory content, or scams.
Pre / Post Conditions	Pre: Job reported or flagged. Post: Job status set to SUSPENDED or ACTIVE, recruiter notified.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Review flagged job details and reported abuse tags from candidates. 2. Administrator clicks 'Suspend Job' and enters violation reason (e.g., Gambling/ Scam). 3. Client dispatches POST /api/v1/admin/jobs/{id}/moderate {action: 'SUSPEND', reason: '...'}. 4. Job Service updates status to SUSPENDED, delists from search index, and invalidates cache. 5. Dispatch warning notification to recruiter and return HTTP 200 OK.
Exception Flows	- [400 Bad Request]: Moderation action missing required reason -> Reject. - [409 Conflict]: Job already deleted or closed by recruiter -> Return status notice.

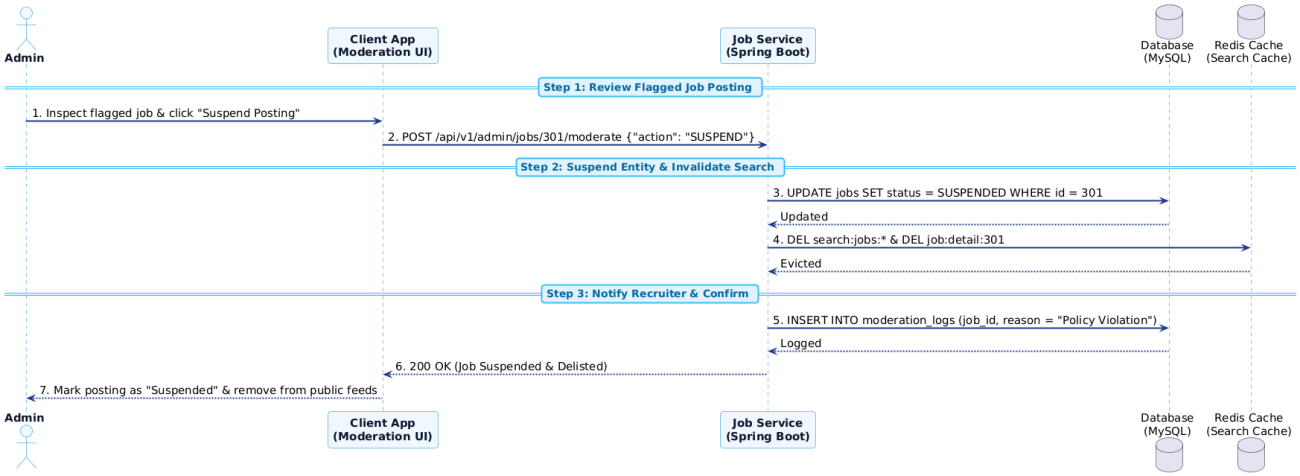


Figure 3.26. Job Posting Moderation & Content Compliance Check Sequence Diagram

4. Administrator Subsystem: 4.5. SePay VietQR Payment Reconciliation & Refund Processing

Use Case Name	UC-27: SePay VietQR Payment Reconciliation & Refund
Actors	System Administrator
Description	Resolves disputed transactions, manual payment reconciliations with SePay banking gateway, and refunds.
Pre / Post Conditions	Pre: Disputed transaction. Post: Transaction reconciled, invoice updated, quota corrected.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Search disputed order by bank transfer reference code (e.g. 'FT26098231'). 2. Client dispatches GET /api/v1/admin/payments/reconcile?ref=FT26098231. 3. Payment Service queries SePay API for authentic bank statement transaction details. 4. Cross-match bank amount with order invoice; manually mark order as COMPLETED. 5. Credit recruiter quota and log manual financial adjustment in payment_audits table.
Exception Flows	- [404 Not Found]: Bank reference not found on SePay gateway -> Require manual statement review. - [400 Bad Request]: Transfer amount less than package invoice price -> Flag underpayment.

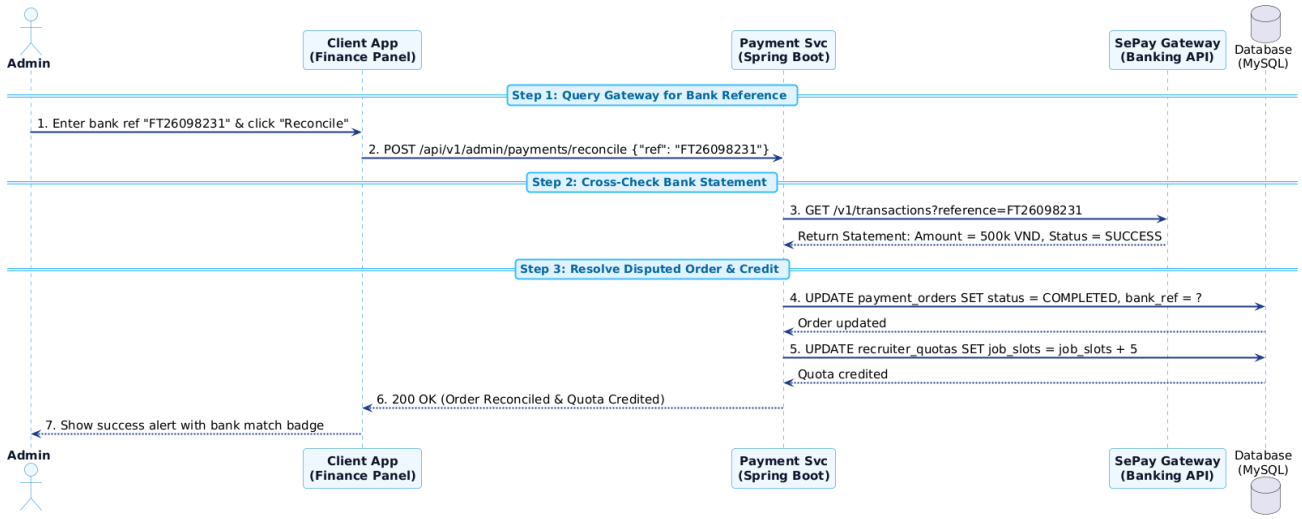


Figure 3.27. SePay VietQR Payment Reconciliation & Refund Processing Sequence Diagram

4. Administrator Subsystem: 4.6. Skill Taxonomy & System Catalog Management

Use Case Name	UC-28: Skill Taxonomy & System Catalog Management
Actors	System Administrator
Description	Maintains standardized IT skill taxonomies, categories (Backend, Frontend, AI/ML), and synonyms.
Pre / Post Conditions	Pre: Admin logged in. Post: New standardized skill persisted, skill autocomplete cache rebuilt.
Flow of Events (Actor Actions & System Responses)	
Basic Flow	1. Enter new skill name ('NestJS'), category ('Backend Framework'), and synonyms ('Nest.js'). 2. Client dispatches POST /api/v1/admin/skills. 3. Taxonomy Service validates skill name uniqueness in skills table. 4. Insert skill record, map to category ID, and rebuild Redis trie/autocomplete index. 5. Return HTTP 201 Created with new skill taxonomy ID.
Exception Flows	- [409 Conflict]: Skill name already registered in taxonomy -> Return duplicate error. - [400 Bad Request]: Empty skill name or invalid category ID -> Reject creation.

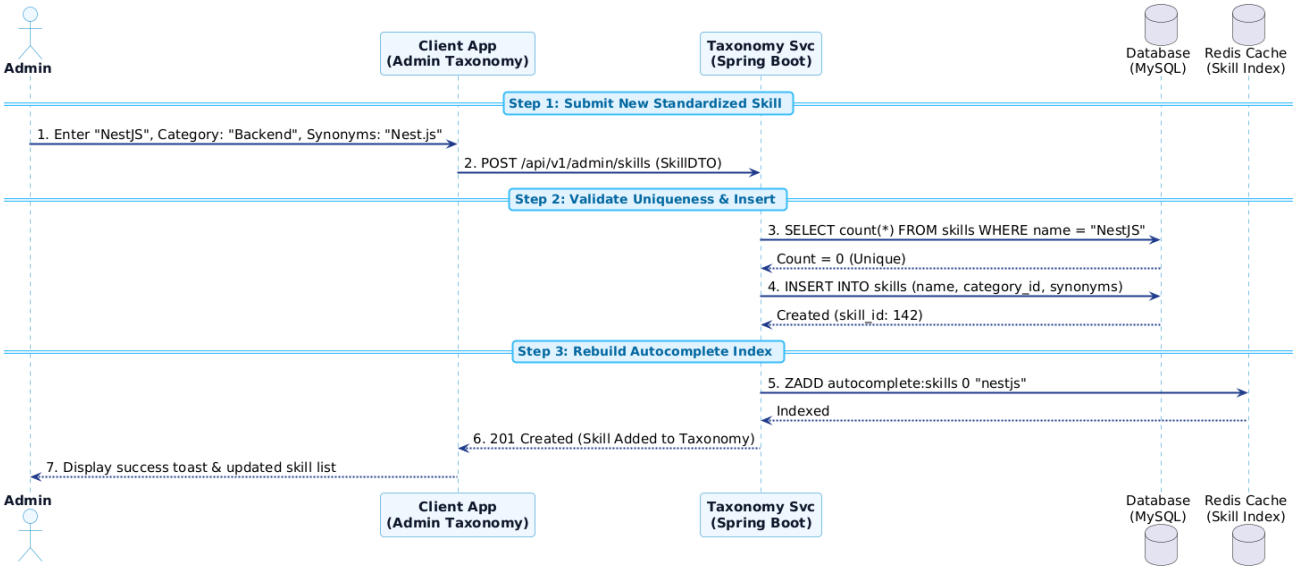


Figure 3.28. Skill Taxonomy & System Catalog Management Sequence Diagram

IV. Class Diagrams

1. Identity, Recruitment & Job Catalog Domain Model

The domain class diagram below models the core recruitment catalog and identity entities, illustrating relationships among user accounts, candidate profiles, verified company entities, job postings, and skill taxonomies.

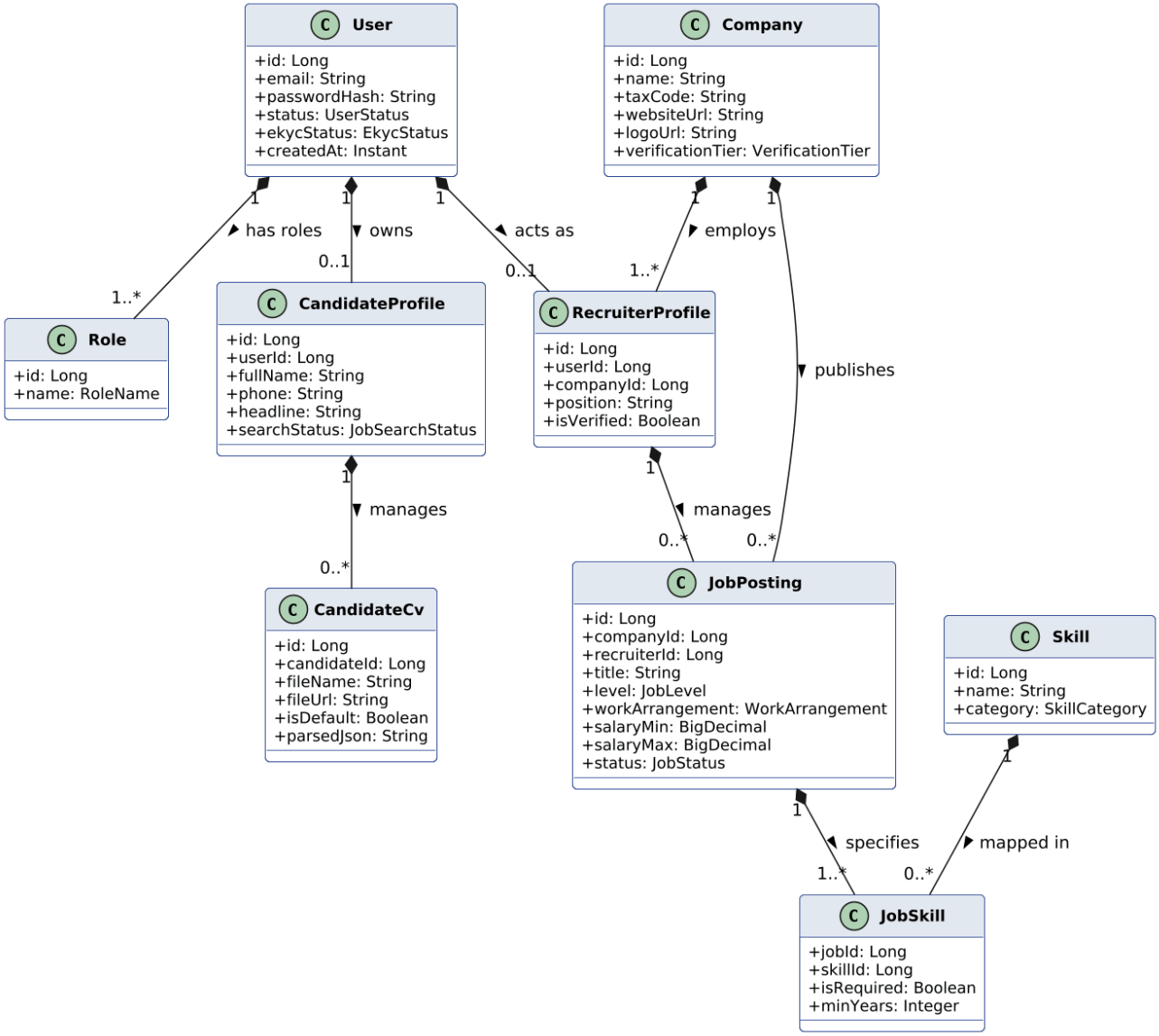


Figure 4.1. Identity, Recruitment and Job Catalog Class Diagram

2. Application Pipeline, AI Assessment & Financial Settlement Domain Model

The domain class diagram below illustrates the operational lifecycle of candidate applications, automated AI match evaluation, interactive mock interview turns, and VietQR financial settlement.

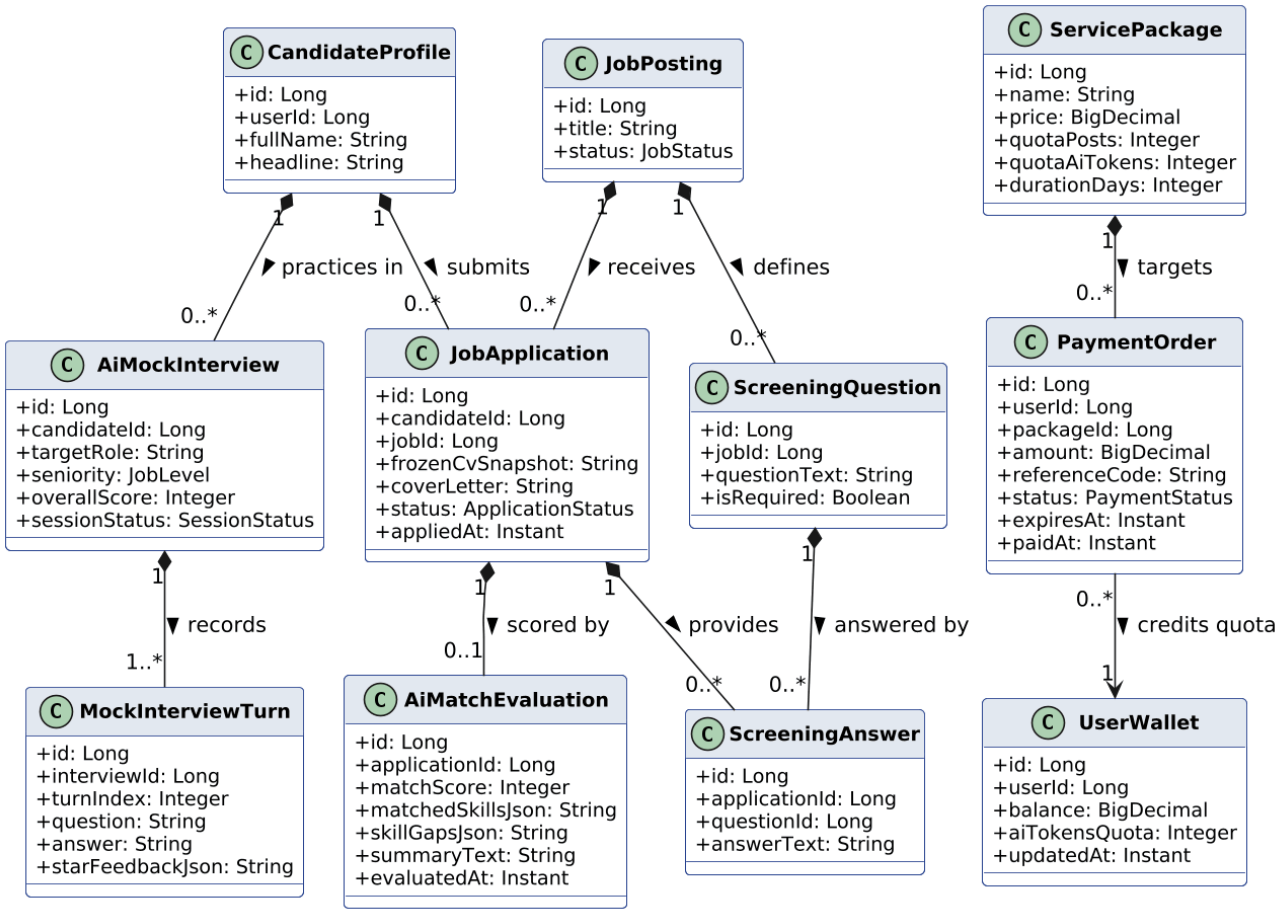


Figure 4.2. Application Pipeline, AI Assessment and Financial Settlement Class Diagram

V. Data Flow Diagrams (DFD)

1. Level-0 Context Diagram

The Level-0 Context Diagram delineates the boundary of the InterVue platform, modeling external data exchanges across Candidates, Recruiters, Administrators, FPT.AI eKYC Gateway, Gemini AI Engine, and SePay Payment Service.

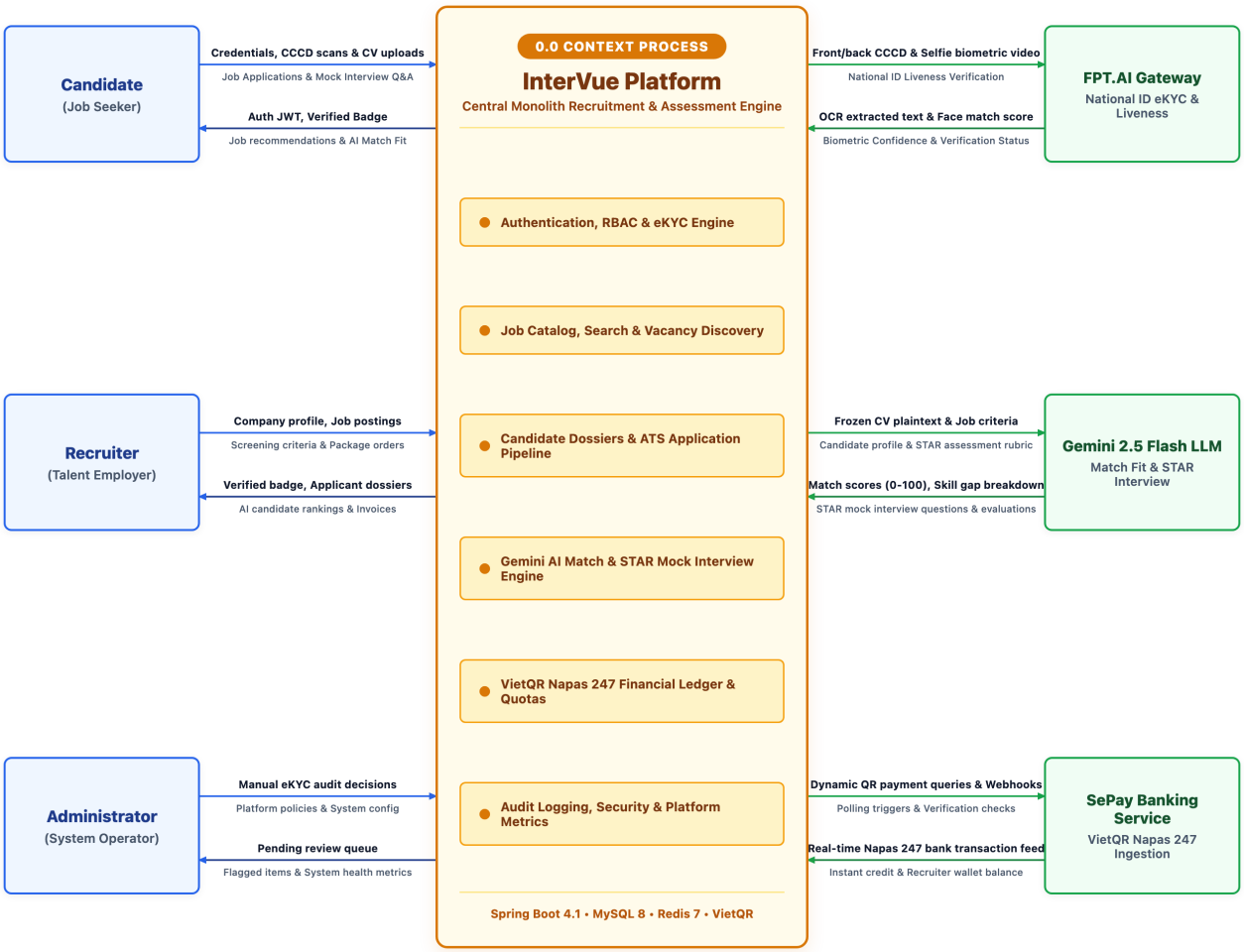


Figure 5.1. Level-0 Context Data Flow Diagram of InterVue Platform

2. Level-1 Data Flow Diagram

The Level-1 Data Flow Diagram decomposes the platform into five core operational processes: Authentication & Identity, eKYC Verification, Job & Application Pipeline, AI Assessment Engine, and Payment & Billing.

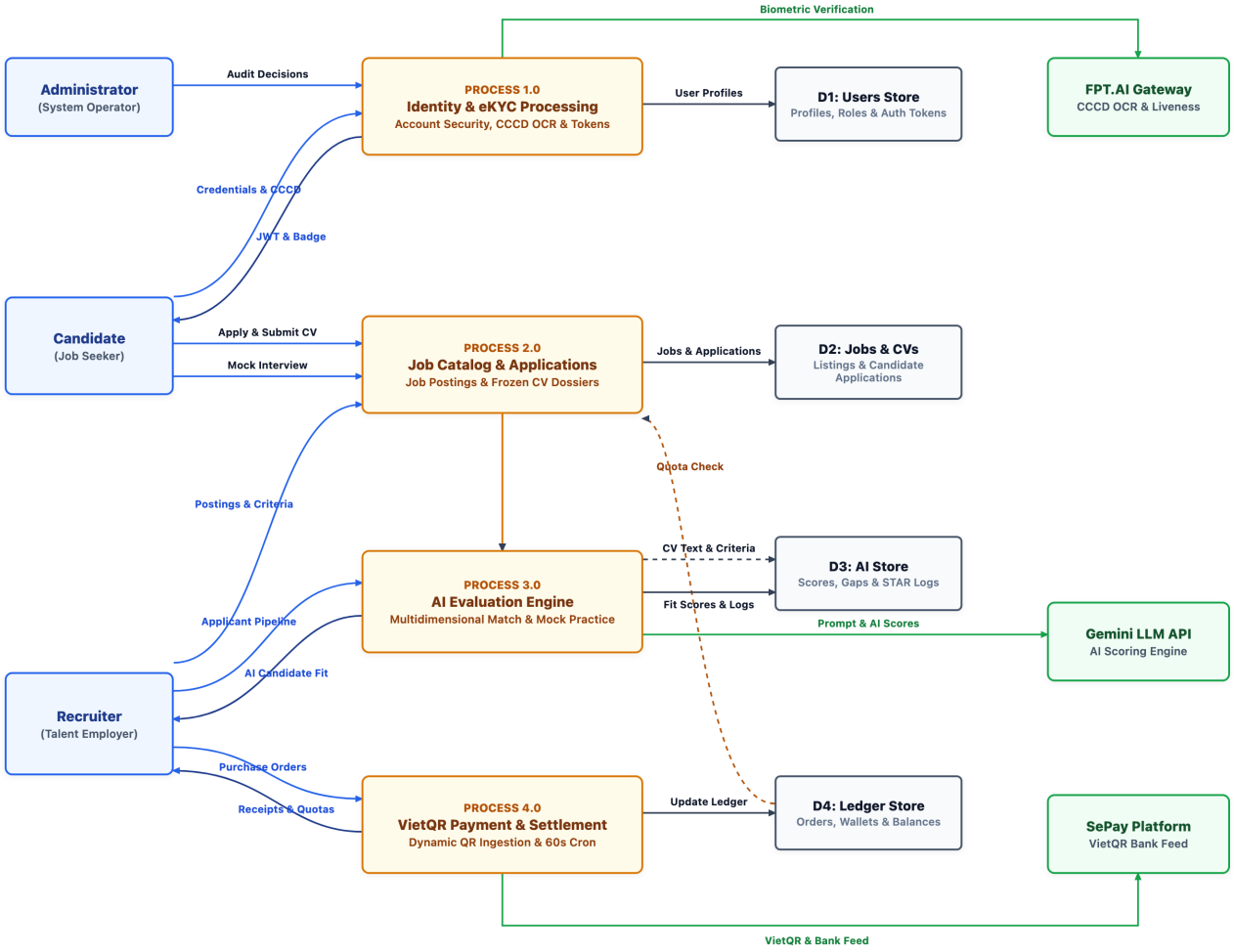


Figure 5.2. Level-1 Data Flow Diagram of InterVue System Processes

VI. Database Design

1. Entity Relationship Diagram

The Entity Relationship Diagram provides a comprehensive conceptual and physical relational model of the InterVue platform, mapping all sixteen persistent data entities across identity management, candidate dossiers, corporate recruitment pipelines, AI evaluation engines, and VietQR financial settlement.

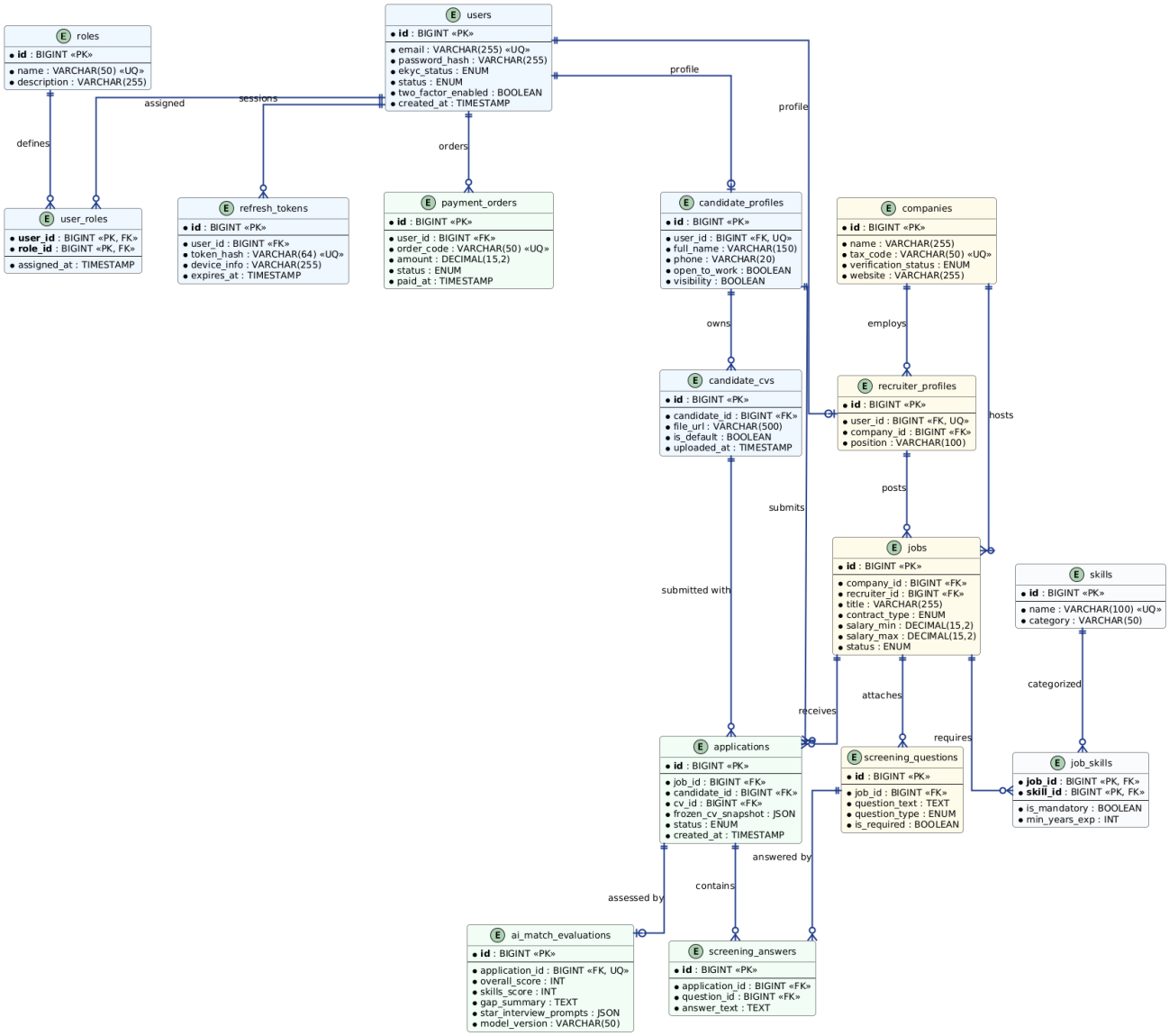


Figure 6.1. Comprehensive Entity Relationship Diagram (ERD) of InterVue Platform

2. Schema Overview

The schema overview categorizes all sixteen relational database tables implemented on MySQL 8.4 utilizing InnoDB storage engines and utf8mb4 character collation, outlining the primary functional scope and storage responsibility of each relational entity.

STT	Name	Description
1	**roles**	Stores master system roles for Spring Security RBAC authorization (CANDIDATE, RECRUITER, ADMIN)
2	**users**	Manages central user authentication accounts, eKYC status, 2FA settings, and soft deletion
3	**user_roles**	Associative join entity enabling many-to-many role assignments per user account
4	**refresh_tokens**	Tracks opaque hashed JWT refresh tokens, device fingerprints, and session expiry
5	**candidate_profiles**	Holds candidate profile info, contact details, career headline, and job search visibility
6	**candidate_cvs**	Manages uploaded CV document files, cloud storage URLs, and default resume flags
7	**companies**	Stores corporate employer entities, tax identification codes, and verification status
8	**recruiter_profiles**	Associates enterprise recruitment staff accounts with verified corporate employers
9	**jobs**	Manages job vacancy postings, contract types, work arrangements, salary bands, and lifecycle
10	**skills**	Master catalog of standardized IT technical skills and competencies
11	**job_skills**	Maps mandatory and optional technical skill requirements to job vacancies
12	**screening_questions**	Custom screening questions attached to jobs (Text, Single Choice, Multiple Choice)
13	**applications**	Candidate job application submissions containing frozen CV snapshots and pipeline states
14	**screening_answers**	Applicant responses to employer screening questions attached to applications
15	**ai_match_evaluations**	AI-generated compatibility scores (0–100), skill GAP analysis, and STAR interview prompts
16	**payment_orders**	Financial transaction ledger for VietQR recruitment service package purchases

Table 6.1. Relational Database Schema Overview of InterVue Platform

3. Detail Schema

The detailed schema specification defines the complete physical field catalog, data types, constraints, and operational logic for all relational database entities.

3.1. roles

roles

No.	Key	Nullable	Field name	Type	Reference Table	Description
1	**PK**	No	**id**	BIGINT		Primary key surrogate identifier of system role
2		No	**name**	VARCHAR(50)		Unique role name identifier (CANDIDATE, RECRUITER, ADMIN)
3		Yes	**description**	VARCHAR(255)		Human-readable description of role permissions

3.2. users

users

No.	Key	Nullable	Field name	Type	Reference Table	Description
1	**PK**	No	**id**	BIGINT		Unique account identifier
2		No	**email**	VARCHAR(255)		Unique user email address used for login and notification
3		No	**password_hash**	VARCHAR(255)		Salted BCrypt cryptographic hash of user password
4		No	**status**	ENUM		Account state: ACTIVE, INACTIVE, BANNED
5		No	**ekyc_status**	ENUM		Identity state: NOT_SUBMITTED, PENDING, APPROVED, REJECTED
6		No	**is_email_verified**	BOOLEAN		Email verification token confirmation flag
7		No	**two_factor_enabled**	BOOLEAN		TOTP two-factor authentication activation toggle
8		No	**created_at**	TIMESTAMP		UTC record creation timestamp
9		No	**updated_at**	TIMESTAMP		UTC record last update timestamp
10		Yes	**deleted_at**	TIMESTAMP		UTC soft-deletion timestamp for reversible account removal

3.3. user_roles

user_roles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK, FK**	No	**user_id**	BIGINT	users(id)	Reference to user account holding assigned role
2	**PK, FK**	No	**role_id**	BIGINT	roles(id)	Reference to granted system authority role
3		No	**assigned_at**	TIMESTAMP		UTC timestamp when role was granted to account

3.4. refresh_tokens

refresh_tokens						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary surrogate key of session record
2	**FK**	No	**user_id**	BIGINT	users(id)	Reference to authenticated user account
3		No	**token_hash**	VARCHAR(64)		SHA-256 hash of opaque refresh token cookie
4		Yes	**device_info**	VARCHAR(255)		Client browser agent and IP telemetry metadata
5		No	**expires_at**	TIMESTAMP		Absolute expiration timestamp of token family
6		Yes	**revoked_at**	TIMESTAMP		Revocation timestamp if manually logged out or rotated
7		No	**created_at**	TIMESTAMP		UTC session establishment timestamp

3.5. candidate_profiles

candidate_profiles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary key of candidate profile record
2	**FK**	No	**user_id**	BIGINT	users(id)	One-to-one foreign key reference to user account
3		No	**full_name**	VARCHAR(150)		Full legal candidate name verified via eKYC CCCD
4		Yes	**phone**	VARCHAR(20)		Candidate contact telephone number
5		Yes	**headline**	VARCHAR(255)		Professional career summary or headline title
6		Yes	**bio**	TEXT		Detailed autobiographical background and intro
7		No	**open_to_work**	BOOLEAN		Flag indicating candidate is actively seeking employment
8		No	**visibility**	BOOLEAN		Privacy toggle permitting profile discovery by recruiters
9		No	**created_at**	TIMESTAMP		UTC profile creation timestamp
10		No	**updated_at**	TIMESTAMP		UTC profile last modification timestamp

3.6. candidate_cvs

candidate_cvs						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique CV asset identifier
2	**FK**	No	**candidate_id**	BIGINT	candidate_profiles(id)	Reference to owner candidate profile
3		No	**file_url**	VARCHAR(500)		Secure cloud storage object key or presigned URL
4		No	**file_name**	VARCHAR(255)		Original file name uploaded by candidate
5		No	**is_default**	BOOLEAN		Flag designating CV as primary default application resume
6		No	**uploaded_at**	TIMESTAMP		UTC upload timestamp

3.7. companies

companies						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary corporate employer identifier
2		No	**name**	VARCHAR(255)		Official registered legal name of enterprise
3		No	**tax_code**	VARCHAR(50)		Unique national corporate tax identification code
4		Yes	**website**	VARCHAR(255)		Official enterprise corporate web URL
5		Yes	**logo_url**	VARCHAR(500)		Cloud storage key for enterprise logo branding image
6		No	**verification_status**	ENUM		Audit state: UNVERIFIED, PENDING, VERIFIED, REJECTED
7		No	**created_at**	TIMESTAMP		UTC enterprise registration timestamp
8		No	**updated_at**	TIMESTAMP		UTC enterprise verification update timestamp

3.8. recruiter_profiles

recruiter_profiles						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique recruiter personnel profile identifier
2	**FK**	No	**user_id**	BIGINT	users(id)	One-to-one reference to recruiter user account
3	**FK**	No	**company_id**	BIGINT	companies(id)	Reference to corporate employer represented by recruiter
4		Yes	**position**	VARCHAR(100)		Corporate job title (e.g., HR Manager, Tech Recruiter)
5		No	**created_at**	TIMESTAMP		UTC recruiter profile creation timestamp
6		No	**updated_at**	TIMESTAMP		UTC recruiter profile last update timestamp

3.9. jobs

jobs

No.	Key	Nullable	Field name	Type	Reference Table	Description
1	**PK**	No	**id**	BIGINT		Primary key vacancy posting identifier
2	**FK**	No	**company_id**	BIGINT	companies(id)	Associated verified corporate employer
3	**FK**	No	**recruiter_id**	BIGINT	recruiter_profiles(id)	Recruiter author who published the vacancy
4		No	**title**	VARCHAR(255)		Job opening headline title
5		No	**contract_type**	ENUM		Contract type: FULL_TIME, PART_TIME, INTERNSHIP
6		No	**working_arrangement**	ENUM		Work mode: ONSITE, HYBRID, REMOTE, FLEXIBLE
7		Yes	**salary_min**	DECIMAL(15,2)		Minimum monthly compensation limit in VND
8		Yes	**salary_max**	DECIMAL(15,2)		Maximum monthly compensation limit in VND
9		No	**currency**	VARCHAR(10)		Currency code (Default: VND)
10		No	**status**	ENUM		Lifecycle status: DRAFT, PUBLISHED, CLOSED
11		Yes	**published_at**	TIMESTAMP		Official vacancy publication timestamp
12		No	**created_at**	TIMESTAMP		Record creation timestamp
13		No	**updated_at**	TIMESTAMP		Last modification timestamp

3.10. skills

skills

No.	Key	Nullable	Field name	Type	Reference Table	Description
1	**PK**	No	**id**	BIGINT		Unique skill identifier in taxonomy
2		No	**name**	VARCHAR(100)		Standardized technical skill name (e.g., Java, React, Docker)
3		Yes	**category**	VARCHAR(50)		Technical classification: BACKEND, FRONTEND, DEVOPS, AI
4		No	**created_at**	TIMESTAMP		UTC taxonomy entry creation timestamp

3.11. job_skills

job_skills						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK, FK**	No	**job_id**	BIGINT	jobs(id)	Reference to target job posting
2	**PK, FK**	No	**skill_id**	BIGINT	skills(id)	Reference to standardized taxonomy skill
3		No	**is_mandatory**	BOOLEAN		Flag designating skill as strictly non-negotiable prerequisite
4		Yes	**min_years_exp**	INT		Minimum required years of commercial experience

3.12. screening_questions

screening_questions						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique screening question identifier
2	**FK**	No	**job_id**	BIGINT	jobs(id)	Reference to parent job posting
3		No	**question_text**	TEXT		Screening questionnaire prompt presented to applicant
4		No	**question_type**	ENUM		Input type: TEXT, SINGLE_CHOICE, MULTI_CHOICE
5		No	**is_required**	BOOLEAN		Mandatory answer flag before application submission

3.13. applications

applications						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Primary application transaction identifier
2	**FK**	No	**job_id**	BIGINT	jobs(id)	Target job vacancy opening
3	**FK**	No	**candidate_id**	BIGINT	candidate_profiles(id)	Applying candidate profile
4	**FK**	No	**cv_id**	BIGINT	candidate_cvs(id)	Selected candidate CV asset
5		No	**frozen_cv_snapshot**	JSON		Immutable JSON snapshot of CV attributes captured at apply time
6		No	**status**	ENUM		Pipeline state: APPLIED, IN_REVIEW, INTERVIEW, OFFER, HIRED
7		Yes	**cover_letter**	TEXT		Personal cover letter or AI-assisted pitch statement
8		No	**created_at**	TIMESTAMP		Application submission timestamp
9		No	**updated_at**	TIMESTAMP		Application pipeline stage transition timestamp

3.14. screening_answers

screening_answers						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Unique screening answer record identifier
2	**FK**	No	**application_id**	BIGINT	applications(id)	Associated candidate job application
3	**FK**	No	**question_id**	BIGINT	screening_questions(id)	Corresponding screening question answered
4		No	**answer_text**	TEXT		Candidate response content or selected option values
5		No	**created_at**	TIMESTAMP		UTC answer submission timestamp

3.15. ai_match_evaluations

ai_match_evaluations						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		AI evaluation result record identifier
2	**FK**	No	**application_id**	BIGINT	applications(id)	One-to-one unique link to evaluated job application
3		No	**overall_score**	INT		Overall compatibility rating (0-100) by Gemini LLM
4		No	**skills_score**	INT		Technical skill alignment score (0-100)
5		No	**experience_score**	INT		Seniority and work experience alignment score (0-100)
6		Yes	**strength_summary**	TEXT		Bullet summary of candidate strengths against job posting
7		Yes	**gap_summary**	TEXT		Detailed gap analysis of missing or unmentioned qualifications
8		Yes	**star_interview_prompts**	JSON		Tailored behavioral interview questions targeting GAP areas
9		No	**model_version**	VARCHAR(50)		Active Gemini LLM model identifier for ethical auditability
10		No	**evaluated_at**	TIMESTAMP		UTC evaluation completion timestamp

3.16. payment_orders

payment_orders						
No.	**Key**	**Nullable**	**Field name**	**Type**	**Reference Table**	**Description**
1	**PK**	No	**id**	BIGINT		Financial payment order identifier
2	**FK**	No	**user_id**	BIGINT	users(id)	Purchasing employer or candidate account
3		No	**order_code**	VARCHAR(50)		Unique VietQR transfer reference code (e.g., INV-202610-8832)
4		No	**amount**	DECIMAL(15,2)		Monetary settlement amount in VND currency
5		No	**payment_method**	VARCHAR(50)		Payment gateway method (Default: VIETQR)
6		No	**status**	ENUM		Billing status: PENDING, PAID, EXPIRED, FAILED
7		Yes	**sepay_transaction_id**	VARCHAR(100)		Inbound bank transfer ID reconciled via SePay webhook
8		Yes	**paid_at**	TIMESTAMP		UTC financial confirmation timestamp
9		No	**created_at**	TIMESTAMP		Order initiation timestamp

VII. SiteMap

1. Web SiteMap by Route Group

The Web SiteMap illustrates the hierarchical route organization of the InterVue web portal implemented with Next.js 16 App Router, segregating public traffic, identity flows, candidate workspaces, enterprise recruiter portals, and administrative governance dashboards.

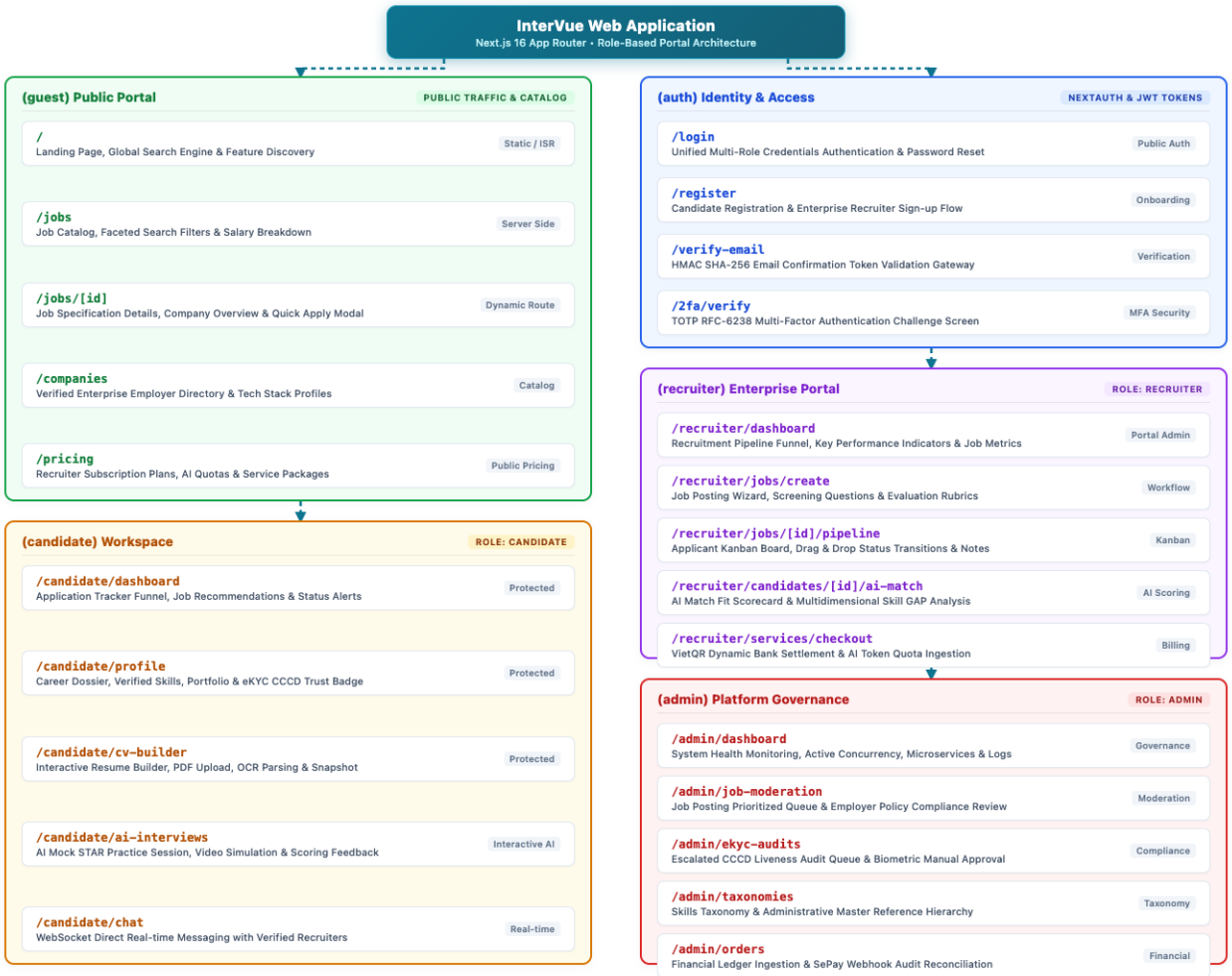


Figure 7.1. Hierarchical Web SiteMap and Route Organization of InterVue Portal

2. Mobile SiteMap & Navigation Architecture

The Mobile SiteMap details the screen navigation topology of the InterVue mobile app developed with Flutter, outlining deep links, guest exploration stacks, biometric authentication screens, and persistent bottom tab navigators.

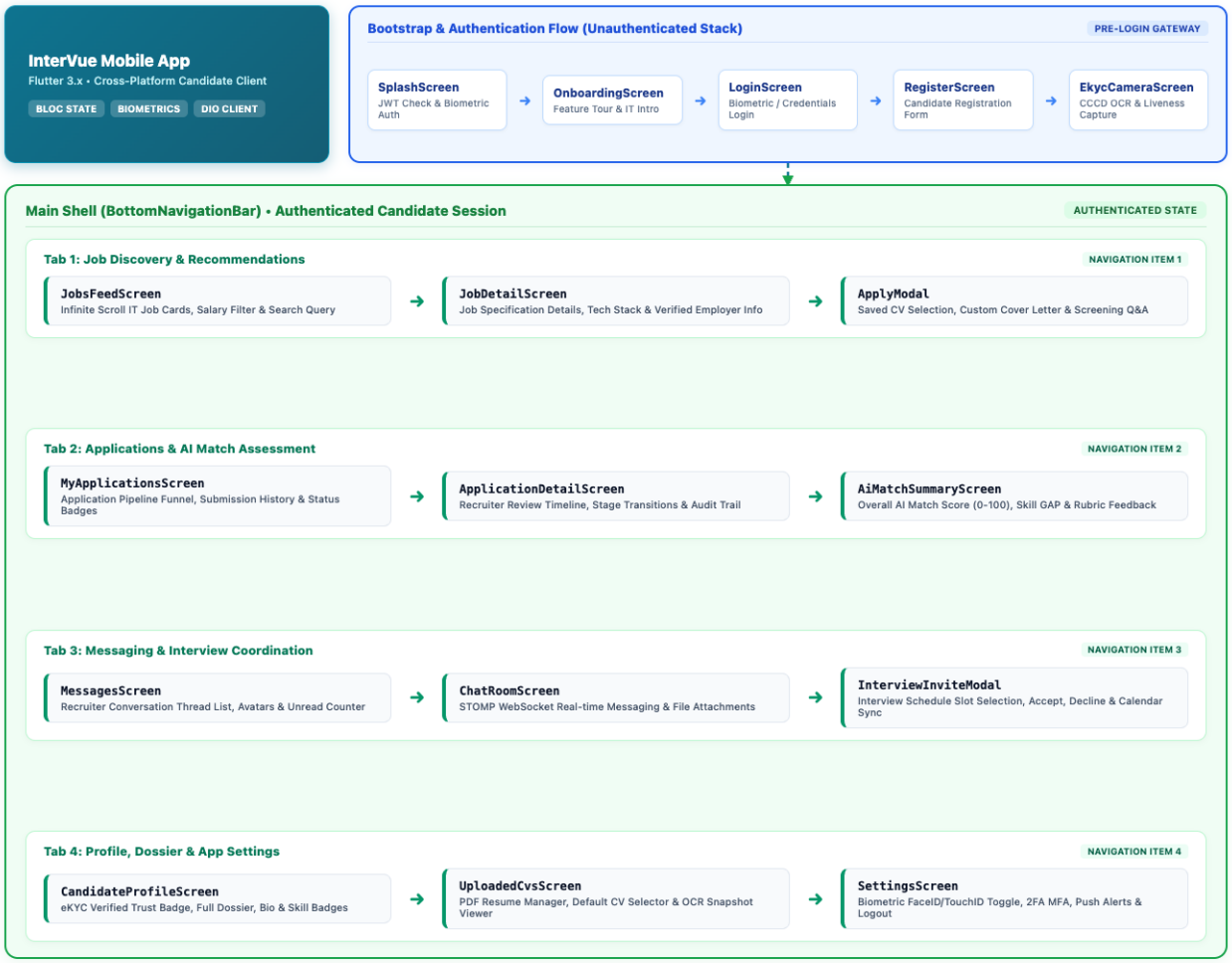


Figure 7.2. Mobile App Screen Flow and Navigation Architecture of InterVue App

Review 03

Acknowledgments and Declaration

Project Documentation for InterVue

Completed on November 11, 2026

The documentation for the **InterVue** project was completed on November 11, 2026. Our team would like to express our sincere gratitude to our supervisor, **Mr. HOÀNG ĐỨC QUANG**, for his continuous guidance, valuable feedback, and encouragement throughout the development of this project.

We are also thankful to our classmates for their constructive input, and to **FPT Aptech** for providing us with the opportunity to apply the knowledge and skills gained during our studies in a practical and meaningful context.

This document is the result of the collective efforts of our team. We confirm that the content presented is original and accurate, with no unauthorized copying or plagiarism. External materials were used for reference only, and any content drafted with the support of digital tools, including artificial intelligence, has been reviewed and verified by our team. We are committed to upholding academic integrity and take full responsibility for this work.

This documentation reflects the dedication and learning process of our team as students still developing our professional skills. We are deeply grateful to our teacher for his support and guidance during this journey.

This is a complete documentation that reflects the dedication, research, and learning process of our team as students who are still developing our professional skills. We sincerely welcome feedback to further improve this

work, and we are deeply grateful to our teacher for his support and guidance during this journey.